

Raw argmax

10 shared seeds: 91001–91010; round 70; validation split (19,867 images)

Table 1: CelebA [raw]: IID / non-IID. COMPLETED VALIDATION COHORT – 10 shared seeds (91001–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	84.72 $\pm$ 11.74	89.57 $\pm$ 0.78	68.19 $\pm$ 5.77	66.01 $\pm$ 3.57	84.37 $\pm$ 2.38	86.85 $\pm$ 5.45	89.51 $\pm$ 1.82	74.76 $\pm$ 13.38	74.51 $\pm$ 13.37	85.48 $\pm$ 2.58
		AEOD $\downarrow$	0.0384 $\pm$ 0.0176	0.0519 $\pm$ 0.0096	0.1004 $\pm$ 0.0410	0.0950 $\pm$ 0.0477	0.0280 $\pm$ 0.0076	0.0428 $\pm$ 0.0137	0.0492 $\pm$ 0.0045	0.0644 $\pm$ 0.1097	0.0522 $\pm$ 0.0913	0.0267 $\pm$ 0.0157
		ASPD $\downarrow$	0.0937 $\pm$ 0.0366	0.1212 $\pm$ 0.0064	0.0993 $\pm$ 0.0380	0.0807 $\pm$ 0.0413	0.0693 $\pm$ 0.0161	0.0958 $\pm$ 0.0353	0.1161 $\pm$ 0.0143	0.0614 $\pm$ 0.0893	0.0591 $\pm$ 0.0756	0.0741 $\pm$ 0.0247
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.47 $\pm$ 0.85	89.79 $\pm$ 0.76	69.47 $\pm$ 4.65	62.59 $\pm$ 5.11	85.55 $\pm$ 2.31	89.59 $\pm$ 0.97	89.83 $\pm$ 0.92	71.45 $\pm$ 11.85	67.06 $\pm$ 11.91	86.18 $\pm$ 2.41
		AEOD $\downarrow$	0.0414 $\pm$ 0.0076	0.0489 $\pm$ 0.0042	0.1158 $\pm$ 0.0635	0.1594 $\pm$ 0.0822	0.0288 $\pm$ 0.0081	0.0388 $\pm$ 0.0057	0.0462 $\pm$ 0.0081	0.0759 $\pm$ 0.0860	0.1307 $\pm$ 0.1184	0.0328 $\pm$ 0.0133
		ASPD $\downarrow$	0.1102 $\pm$ 0.0077	0.1188 $\pm$ 0.0043	0.1120 $\pm$ 0.0554	0.1461 $\pm$ 0.0668	0.0781 $\pm$ 0.0116	0.1092 $\pm$ 0.0071	0.1163 $\pm$ 0.0071	0.0730 $\pm$ 0.0885	0.1171 $\pm$ 0.1067	0.0829 $\pm$ 0.0204
Robust FL	Median	ACC (%) $\uparrow$	86.28 $\pm$ 1.31	86.52 $\pm$ 1.35	84.66 $\pm$ 1.47	85.00 $\pm$ 1.38	85.54 $\pm$ 1.50	86.22 $\pm$ 1.31	86.51 $\pm$ 1.24	84.47 $\pm$ 1.30	84.48 $\pm$ 1.35	85.47 $\pm$ 1.26
		AEOD $\downarrow$	0.0510 $\pm$ 0.0081	0.0610 $\pm$ 0.0087	0.0438 $\pm$ 0.0135	0.0484 $\pm$ 0.0090	0.0534 $\pm$ 0.0096	0.0518 $\pm$ 0.0068	0.0594 $\pm$ 0.0084	0.0425 $\pm$ 0.0118	0.0461 $\pm$ 0.0142	0.0502 $\pm$ 0.0111
		ASPD $\downarrow$	0.0980 $\pm$ 0.0117	0.1062 $\pm$ 0.0111	0.0825 $\pm$ 0.0142	0.0852 $\pm$ 0.0127	0.0965 $\pm$ 0.0124	0.0963 $\pm$ 0.0096	0.1067 $\pm$ 0.0097	0.0799 $\pm$ 0.0113	0.0839 $\pm$ 0.0124	0.0929 $\pm$ 0.0086
Robust FL	FLTrust	ACC (%) $\uparrow$	89.43 $\pm$ 0.75	89.67 $\pm$ 0.82	89.68 $\pm$ 0.74	89.68 $\pm$ 0.74	89.72 $\pm$ 0.64	89.68 $\pm$ 0.60	89.83 $\pm$ 0.69	89.48 $\pm$ 0.70	89.48 $\pm$ 0.70	89.64 $\pm$ 0.71
		AEOD $\downarrow$	0.0432 $\pm$ 0.0075	0.0518 $\pm$ 0.0072	0.0422 $\pm$ 0.0069	0.0422 $\pm$ 0.0069	0.0496 $\pm$ 0.0061	0.0445 $\pm$ 0.0049	0.0493 $\pm$ 0.0054	0.0441 $\pm$ 0.0063	0.0441 $\pm$ 0.0063	0.0475 $\pm$ 0.0052
		ASPD $\downarrow$	0.1111 $\pm$ 0.0062	0.1207 $\pm$ 0.0064	0.1119 $\pm$ 0.0071	0.1119 $\pm$ 0.0071	0.1177 $\pm$ 0.0057	0.1132 $\pm$ 0.0060	0.1197 $\pm$ 0.0044	0.1120 $\pm$ 0.0041	0.1120 $\pm$ 0.0041	0.1153 $\pm$ 0.0053
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.35 $\pm$ 1.33	89.06 $\pm$ 0.92	88.80 $\pm$ 1.06	88.82 $\pm$ 1.21	88.60 $\pm$ 1.12	88.60 $\pm$ 1.14	88.56 $\pm$ 1.19	88.53 $\pm$ 1.31	88.52 $\pm$ 1.60	88.39 $\pm$ 1.14
		AEOD $\downarrow$	0.0455 $\pm$ 0.0116	0.0442 $\pm$ 0.0137	0.0423 $\pm$ 0.0116	0.0385 $\pm$ 0.0131	0.0493 $\pm$ 0.0089	0.0456 $\pm$ 0.0125	0.0491 $\pm$ 0.0132	0.0421 $\pm$ 0.0231	0.0409 $\pm$ 0.0122	0.0597 $\pm$ 0.0189
		ASPD $\downarrow$	0.1083 $\pm$ 0.0100	0.1097 $\pm$ 0.0143	0.1071 $\pm$ 0.0115	0.1042 $\pm$ 0.0152	0.1136 $\pm$ 0.0095	0.1069 $\pm$ 0.0151	0.1110 $\pm$ 0.0103	0.1031 $\pm$ 0.0212	0.1020 $\pm$ 0.0149	0.1188 $\pm$ 0.0190
Robust FL	LASA*	ACC (%) $\uparrow$	89.39 $\pm$ 1.42	89.46 $\pm$ 0.79	85.75 $\pm$ 1.71	85.96 $\pm$ 1.62	88.20 $\pm$ 1.40	89.27 $\pm$ 0.85	89.51 $\pm$ 0.88	85.91 $\pm$ 1.69	85.93 $\pm$ 1.64	88.15 $\pm$ 1.22
		AEOD $\downarrow$	0.0377 $\pm$ 0.0049	0.0493 $\pm$ 0.0066	0.0330 $\pm$ 0.0109	0.0366 $\pm$ 0.0105	0.0450 $\pm$ 0.0073	0.0386 $\pm$ 0.0057	0.0500 $\pm$ 0.0096	0.0301 $\pm$ 0.0069	0.0345 $\pm$ 0.0113	0.0417 $\pm$ 0.0037
		ASPD $\downarrow$	0.1074 $\pm$ 0.0088	0.1176 $\pm$ 0.0059	0.0811 $\pm$ 0.0138	0.0839 $\pm$ 0.0152	0.1050 $\pm$ 0.0113	0.1070 $\pm$ 0.0084	0.1181 $\pm$ 0.0059	0.0767 $\pm$ 0.0129	0.0791 $\pm$ 0.0128	0.1026 $\pm$ 0.0074
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.18 $\pm$ 0.83	89.57 $\pm$ 1.24	49.32 $\pm$ 1.59	50.33 $\pm$ 1.72	83.20 $\pm$ 1.76	89.39 $\pm$ 1.11	89.84 $\pm$ 0.76	59.95 $\pm$ 16.51	61.29 $\pm$ 15.59	78.37 $\pm$ 14.94
		AEOD $\downarrow$	0.0317 $\pm$ 0.0026	0.0369 $\pm$ 0.0071	0.0042 $\pm$ 0.0132	0.0000 $\pm$ 0.0000	0.0205 $\pm$ 0.0098	0.0314 $\pm$ 0.0054	0.0394 $\pm$ 0.0103	0.0066 $\pm$ 0.0094	0.0108 $\pm$ 0.0161	0.0209 $\pm$ 0.0104
		ASPD $\downarrow$	0.0998 $\pm$ 0.0062	0.1078 $\pm$ 0.0092	0.0049 $\pm$ 0.0156	0.0000 $\pm$ 0.0000	0.0491 $\pm$ 0.0199	0.1026 $\pm$ 0.0086	0.1109 $\pm$ 0.0090	0.0169 $\pm$ 0.0251	0.0206 $\pm$ 0.0317	0.0552 $\pm$ 0.0238
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.93 $\pm$ 1.10	90.37 $\pm$ 0.60	77.83 $\pm$ 12.15	75.99 $\pm$ 16.52	86.44 $\pm$ 5.76	89.69 $\pm$ 1.68	90.07 $\pm$ 1.34	81.90 $\pm$ 11.66	80.44 $\pm$ 12.06	79.69 $\pm$ 13.94
		AEOD $\downarrow$	0.0365 $\pm$ 0.0075	0.0363 $\pm$ 0.0086	0.0298 $\pm$ 0.0299	0.0293 $\pm$ 0.0146	0.0368 $\pm$ 0.0144	0.0357 $\pm$ 0.0055	0.0426 $\pm$ 0.0062	0.0400 $\pm$ 0.0214	0.0359 $\pm$ 0.0262	0.0408 $\pm$ 0.0124
		ASPD $\downarrow$	0.1074 $\pm$ 0.0113	0.1118 $\pm$ 0.0100	0.0607 $\pm$ 0.0354	0.0652 $\pm$ 0.0419	0.0903 $\pm$ 0.0325	0.1085 $\pm$ 0.0102	0.1144 $\pm$ 0.0088	0.0851 $\pm$ 0.0350	0.0763 $\pm$ 0.0174	0.0892 $\pm$ 0.0276
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.55 $\pm$ 1.04	88.70 $\pm$ 0.73	88.83 $\pm$ 0.90	89.06 $\pm$ 0.84	88.77 $\pm$ 0.92	88.88 $\pm$ 1.21	88.73 $\pm$ 0.72	88.68 $\pm$ 0.89	88.50 $\pm$ 0.79	88.70 $\pm$ 0.93
		AEOD $\downarrow$	0.0403 $\pm$ 0.0103	0.0368 $\pm$ 0.0134	0.0372 $\pm$ 0.0070	0.0367 $\pm$ 0.0038	0.0347 $\pm$ 0.0139	0.0316 $\pm$ 0.0064	0.0316 $\pm$ 0.0073	0.0353 $\pm$ 0.0082	0.0439 $\pm$ 0.0169	0.0333 $\pm$ 0.0071
		ASPD $\downarrow$	0.1031 $\pm$ 0.0101	0.1008 $\pm$ 0.0103	0.1026 $\pm$ 0.0116	0.1019 $\pm$ 0.0066	0.0998 $\pm$ 0.0106	0.0998 $\pm$ 0.0077	0.0995 $\pm$ 0.0058	0.0989 $\pm$ 0.0112	0.1046 $\pm$ 0.0122	0.0994 $\pm$ 0.0099

All displayed cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
This table uses 886 cu128 and 14 cu130 records; the source cohort contains 14 cu130 records. No 70-round equivalence claim.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Raw view: all nine methods use uncalibrated argmax (margin > 0); same saved checkpoint per ID.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Raw argmax

9 shared seeds: 91002–91010; round 70; validation split (19,867 images)

Table 2: CelebA [raw]: IID / non-IID. COMPLETED VALIDATION COHORT – 9 shared seeds (91002–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	88.39 $\pm$ 1.84	89.72 $\pm$ 0.65	69.08 $\pm$ 5.35	66.61 $\pm$ 3.21	84.91 $\pm$ 1.77	88.09 $\pm$ 4.01	89.91 $\pm$ 1.39	75.58 $\pm$ 13.92	75.31 $\pm$ 13.93	85.69 $\pm$ 2.65
		AEOD $\downarrow$	0.0426 $\pm$ 0.0119	0.0522 $\pm$ 0.0101	0.0939 $\pm$ 0.0376	0.0904 $\pm$ 0.0481	0.0289 $\pm$ 0.0075	0.0457 $\pm$ 0.0107	0.0485 $\pm$ 0.0043	0.0560 $\pm$ 0.1128	0.0510 $\pm$ 0.0968	0.0273 $\pm$ 0.0165
		ASPD $\downarrow$	0.1041 $\pm$ 0.0171	0.1225 $\pm$ 0.0053	0.0962 $\pm$ 0.0389	0.0781 $\pm$ 0.0429	0.0702 $\pm$ 0.0168	0.1050 $\pm$ 0.0215	0.1190 $\pm$ 0.0117	0.0552 $\pm$ 0.0923	0.0603 $\pm$ 0.0801	0.0748 $\pm$ 0.0260
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.61 $\pm$ 0.78	89.91 $\pm$ 0.71	70.47 $\pm$ 3.62	63.49 $\pm$ 4.50	86.17 $\pm$ 1.29	89.76 $\pm$ 0.83	89.92 $\pm$ 0.92	72.23 $\pm$ 12.29	67.94 $\pm$ 12.28	86.50 $\pm$ 2.31
		AEOD $\downarrow$	0.0411 $\pm$ 0.0080	0.0483 $\pm$ 0.0040	0.1030 $\pm$ 0.0521	0.1435 $\pm$ 0.0691	0.0301 $\pm$ 0.0076	0.0380 $\pm$ 0.0054	0.0448 $\pm$ 0.0073	0.0688 $\pm$ 0.0881	0.1327 $\pm$ 0.1254	0.0318 $\pm$ 0.0137
		ASPD $\downarrow$	0.1108 $\pm$ 0.0078	0.1189 $\pm$ 0.0046	0.1009 $\pm$ 0.0454	0.1346 $\pm$ 0.0595	0.0797 $\pm$ 0.0111	0.1095 $\pm$ 0.0074	0.1158 $\pm$ 0.0073	0.0676 $\pm$ 0.0922	0.1180 $\pm$ 0.1131	0.0838 $\pm$ 0.0214
Robust FL	Median	ACC (%) $\uparrow$	86.51 $\pm$ 1.16	86.71 $\pm$ 1.30	84.87 $\pm$ 1.39	85.25 $\pm$ 1.20	85.73 $\pm$ 1.47	86.40 $\pm$ 1.26	86.51 $\pm$ 1.31	84.54 $\pm$ 1.36	84.59 $\pm$ 1.39	85.54 $\pm$ 1.32
		AEOD $\downarrow$	0.0509 $\pm$ 0.0086	0.0610 $\pm$ 0.0093	0.0433 $\pm$ 0.0142	0.0483 $\pm$ 0.0096	0.0532 $\pm$ 0.0101	0.0510 $\pm$ 0.0066	0.0607 $\pm$ 0.0078	0.0425 $\pm$ 0.0125	0.0448 $\pm$ 0.0144	0.0503 $\pm$ 0.0118
		ASPD $\downarrow$	0.0988 $\pm$ 0.0121	0.1070 $\pm$ 0.0114	0.0830 $\pm$ 0.0150	0.0860 $\pm$ 0.0132	0.0974 $\pm$ 0.0128	0.0965 $\pm$ 0.0102	0.1074 $\pm$ 0.0100	0.0800 $\pm$ 0.0120	0.0833 $\pm$ 0.0130	0.0930 $\pm$ 0.0091
Robust FL	FLTrust	ACC (%) $\uparrow$	89.47 $\pm$ 0.78	89.73 $\pm$ 0.85	89.73 $\pm$ 0.76	89.73 $\pm$ 0.76	89.79 $\pm$ 0.64	89.77 $\pm$ 0.56	89.87 $\pm$ 0.72	89.55 $\pm$ 0.70	89.55 $\pm$ 0.70	89.68 $\pm$ 0.74
		AEOD $\downarrow$	0.0420 $\pm$ 0.0069	0.0521 $\pm$ 0.0075	0.0416 $\pm$ 0.0071	0.0416 $\pm$ 0.0071	0.0494 $\pm$ 0.0064	0.0450 $\pm$ 0.0049	0.0488 $\pm$ 0.0056	0.0441 $\pm$ 0.0066	0.0441 $\pm$ 0.0066	0.0481 $\pm$ 0.0051
		ASPD $\downarrow$	0.1108 $\pm$ 0.0065	0.1214 $\pm$ 0.0064	0.1124 $\pm$ 0.0074	0.1124 $\pm$ 0.0074	0.1181 $\pm$ 0.0059	0.1142 $\pm$ 0.0055	0.1194 $\pm$ 0.0046	0.1124 $\pm$ 0.0041	0.1124 $\pm$ 0.0041	0.1161 $\pm$ 0.0049
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.58 $\pm$ 1.18	89.24 $\pm$ 0.77	88.85 $\pm$ 1.11	88.90 $\pm$ 1.26	88.86 $\pm$ 0.81	88.67 $\pm$ 1.19	88.73 $\pm$ 1.13	88.74 $\pm$ 1.21	88.89 $\pm$ 1.17	88.46 $\pm$ 1.18
		AEOD $\downarrow$	0.0446 $\pm$ 0.0120	0.0455 $\pm$ 0.0139	0.0428 $\pm$ 0.0122	0.0363 $\pm$ 0.0116	0.0478 $\pm$ 0.0078	0.0464 $\pm$ 0.0130	0.0490 $\pm$ 0.0140	0.0451 $\pm$ 0.0223	0.0421 $\pm$ 0.0123	0.0606 $\pm$ 0.0198
		ASPD $\downarrow$	0.1093 $\pm$ 0.0101	0.1124 $\pm$ 0.0121	0.1076 $\pm$ 0.0121	0.1032 $\pm$ 0.0158	0.1136 $\pm$ 0.0101	0.1072 $\pm$ 0.0159	0.1117 $\pm$ 0.0106	0.1077 $\pm$ 0.0166	0.1056 $\pm$ 0.0100	0.1198 $\pm$ 0.0199
Robust FL	LASA*	ACC (%) $\uparrow$	89.77 $\pm$ 0.81	89.59 $\pm$ 0.72	86.15 $\pm$ 1.21	86.15 $\pm$ 1.61	88.38 $\pm$ 1.35	89.46 $\pm$ 0.63	89.68 $\pm$ 0.72	86.33 $\pm$ 1.10	86.32 $\pm$ 1.16	88.34 $\pm$ 1.13
		AEOD $\downarrow$	0.0373 $\pm$ 0.0050	0.0489 $\pm$ 0.0069	0.0315 $\pm$ 0.0103	0.0379 $\pm$ 0.0103	0.0446 $\pm$ 0.0077	0.0379 $\pm$ 0.0056	0.0485 $\pm$ 0.0090	0.0301 $\pm$ 0.0074	0.0350 $\pm$ 0.0118	0.0409 $\pm$ 0.0028
		ASPD $\downarrow$	0.1099 $\pm$ 0.0042	0.1184 $\pm$ 0.0057	0.0810 $\pm$ 0.0146	0.0861 $\pm$ 0.0143	0.1059 $\pm$ 0.0116	0.1078 $\pm$ 0.0085	0.1179 $\pm$ 0.0062	0.0782 $\pm$ 0.0127	0.0818 $\pm$ 0.0101	0.1033 $\pm$ 0.0075
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.27 $\pm$ 0.84	89.65 $\pm$ 1.29	49.42 $\pm$ 1.64	50.56 $\pm$ 1.67	83.20 $\pm$ 1.87	89.52 $\pm$ 1.09	89.95 $\pm$ 0.71	61.24 $\pm$ 16.96	62.72 $\pm$ 15.82	77.66 $\pm$ 15.66
		AEOD $\downarrow$	0.0316 $\pm$ 0.0028	0.0369 $\pm$ 0.0075	0.0046 $\pm$ 0.0139	0.0000 $\pm$ 0.0000	0.0185 $\pm$ 0.0080	0.0301 $\pm$ 0.0039	0.0380 $\pm$ 0.0099	0.0073 $\pm$ 0.0097	0.0119 $\pm$ 0.0166	0.0188 $\pm$ 0.0084
		ASPD $\downarrow$	0.1002 $\pm$ 0.0064	0.1085 $\pm$ 0.0094	0.0055 $\pm$ 0.0164	0.0000 $\pm$ 0.0000	0.0471 $\pm$ 0.0199	0.1023 $\pm$ 0.0091	0.1104 $\pm$ 0.0094	0.0188 $\pm$ 0.0259	0.0226 $\pm$ 0.0329	0.0523 $\pm$ 0.0233
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	90.22 $\pm$ 0.65	90.44 $\pm$ 0.59	76.75 $\pm$ 12.37	74.59 $\pm$ 16.88	86.38 $\pm$ 6.11	90.20 $\pm$ 0.46	90.17 $\pm$ 1.38	81.41 $\pm$ 12.25	79.86 $\pm$ 12.64	78.76 $\pm$ 14.45
		AEOD $\downarrow$	0.0371 $\pm$ 0.0076	0.0362 $\pm$ 0.0091	0.0308 $\pm$ 0.0315	0.0298 $\pm$ 0.0154	0.0367 $\pm$ 0.0153	0.0359 $\pm$ 0.0058	0.0415 $\pm$ 0.0055	0.0407 $\pm$ 0.0225	0.0371 $\pm$ 0.0275	0.0404 $\pm$ 0.0131
		ASPD $\downarrow$	0.1101 $\pm$ 0.0078	0.1131 $\pm$ 0.0098	0.0588 $\pm$ 0.0369	0.0627 $\pm$ 0.0437	0.0903 $\pm$ 0.0345	0.1114 $\pm$ 0.0045	0.1154 $\pm$ 0.0087	0.0859 $\pm$ 0.0370	0.0780 $\pm$ 0.0177	0.0882 $\pm$ 0.0291
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.65 $\pm$ 1.05	88.73 $\pm$ 0.77	89.00 $\pm$ 0.77	89.23 $\pm$ 0.67	88.92 $\pm$ 0.82	88.88 $\pm$ 1.28	88.81 $\pm$ 0.71	88.67 $\pm$ 0.95	88.51 $\pm$ 0.84	88.81 $\pm$ 0.92
		AEOD $\downarrow$	0.0400 $\pm$ 0.0109	0.0377 $\pm$ 0.0140	0.0380 $\pm$ 0.0069	0.0365 $\pm$ 0.0039	0.0322 $\pm$ 0.0123	0.0301 $\pm$ 0.0045	0.0313 $\pm$ 0.0077	0.0356 $\pm$ 0.0087	0.0451 $\pm$ 0.0175	0.0329 $\pm$ 0.0074
		ASPD $\downarrow$	0.1034 $\pm$ 0.0106	0.1020 $\pm$ 0.0102	0.1045 $\pm$ 0.0103	0.1034 $\pm$ 0.0050	0.0997 $\pm$ 0.0112	0.0991 $\pm$ 0.0079	0.0998 $\pm$ 0.0060	0.0994 $\pm$ 0.0118	0.1058 $\pm$ 0.0123	0.1002 $\pm$ 0.0102

All displayed cells use the same 9 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Raw view: all nine methods use uncalibrated argmax (margin > 0); same saved checkpoint per ID.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Raw argmax

6 shared seeds: 91005–91010; round 70; validation split (19,867 images)

Table 3: CelebA [raw]: IID / non-IID. COMPLETED VALIDATION COHORT – 6 shared seeds (91005–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	87.99 $\pm$ 2.11	89.71 $\pm$ 0.80	69.36 $\pm$ 6.14	67.41 $\pm$ 3.17	85.70 $\pm$ 1.38	86.90 $\pm$ 4.54	89.58 $\pm$ 1.62	78.23 $\pm$ 12.19	78.69 $\pm$ 12.03	86.13 $\pm$ 3.06
		AEOD $\downarrow$	0.0406 $\pm$ 0.0131	0.0535 $\pm$ 0.0119	0.0852 $\pm$ 0.0377	0.0823 $\pm$ 0.0434	0.0280 $\pm$ 0.0074	0.0464 $\pm$ 0.0125	0.0479 $\pm$ 0.0036	0.0756 $\pm$ 0.1376	0.0652 $\pm$ 0.1170	0.0312 $\pm$ 0.0166
		ASPD $\downarrow$	0.0992 $\pm$ 0.0184	0.1227 $\pm$ 0.0063	0.0866 $\pm$ 0.0413	0.0698 $\pm$ 0.0304	0.0738 $\pm$ 0.0142	0.0977 $\pm$ 0.0234	0.1151 $\pm$ 0.0128	0.0788 $\pm$ 0.1078	0.0804 $\pm$ 0.0927	0.0817 $\pm$ 0.0283
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.39 $\pm$ 0.89	89.65 $\pm$ 0.75	70.49 $\pm$ 3.83	65.00 $\pm$ 4.85	85.87 $\pm$ 1.25	89.56 $\pm$ 0.96	89.73 $\pm$ 1.11	75.25 $\pm$ 10.47	70.49 $\pm$ 12.37	86.17 $\pm$ 2.80
		AEOD $\downarrow$	0.0414 $\pm$ 0.0096	0.0480 $\pm$ 0.0047	0.0912 $\pm$ 0.0604	0.1179 $\pm$ 0.0658	0.0305 $\pm$ 0.0060	0.0388 $\pm$ 0.0033	0.0431 $\pm$ 0.0055	0.0727 $\pm$ 0.1009	0.0952 $\pm$ 0.1115	0.0299 $\pm$ 0.0162
		ASPD $\downarrow$	0.1090 $\pm$ 0.0082	0.1170 $\pm$ 0.0044	0.0961 $\pm$ 0.0536	0.1133 $\pm$ 0.0622	0.0790 $\pm$ 0.0076	0.1082 $\pm$ 0.0085	0.1133 $\pm$ 0.0064	0.0770 $\pm$ 0.1069	0.0879 $\pm$ 0.1083	0.0814 $\pm$ 0.0256
Robust FL	Median	ACC (%) $\uparrow$	86.37 $\pm$ 1.01	86.58 $\pm$ 1.26	84.48 $\pm$ 0.83	84.83 $\pm$ 0.80	85.45 $\pm$ 1.23	86.19 $\pm$ 1.06	86.20 $\pm$ 1.30	84.49 $\pm$ 1.15	84.57 $\pm$ 1.19	85.68 $\pm$ 1.19
		AEOD $\downarrow$	0.0504 $\pm$ 0.0088	0.0602 $\pm$ 0.0102	0.0402 $\pm$ 0.0154	0.0440 $\pm$ 0.0083	0.0526 $\pm$ 0.0099	0.0525 $\pm$ 0.0042	0.0579 $\pm$ 0.0076	0.0423 $\pm$ 0.0100	0.0433 $\pm$ 0.0112	0.0508 $\pm$ 0.0143
		ASPD $\downarrow$	0.0977 $\pm$ 0.0107	0.1054 $\pm$ 0.0109	0.0786 $\pm$ 0.0124	0.0807 $\pm$ 0.0088	0.0956 $\pm$ 0.0101	0.0969 $\pm$ 0.0033	0.1044 $\pm$ 0.0095	0.0786 $\pm$ 0.0089	0.0817 $\pm$ 0.0094	0.0950 $\pm$ 0.0072
Robust FL	FLTrust	ACC (%) $\uparrow$	89.41 $\pm$ 0.86	89.73 $\pm$ 0.92	89.61 $\pm$ 0.88	89.61 $\pm$ 0.88	89.67 $\pm$ 0.71	89.60 $\pm$ 0.61	89.75 $\pm$ 0.87	89.48 $\pm$ 0.63	89.48 $\pm$ 0.63	89.66 $\pm$ 0.82
		AEOD $\downarrow$	0.0392 $\pm$ 0.0056	0.0488 $\pm$ 0.0070	0.0385 $\pm$ 0.0059	0.0385 $\pm$ 0.0059	0.0475 $\pm$ 0.0063	0.0428 $\pm$ 0.0041	0.0482 $\pm$ 0.0064	0.0435 $\pm$ 0.0048	0.0435 $\pm$ 0.0048	0.0473 $\pm$ 0.0034
		ASPD $\downarrow$	0.1087 $\pm$ 0.0063	0.1189 $\pm$ 0.0046	0.1088 $\pm$ 0.0048	0.1088 $\pm$ 0.0048	0.1151 $\pm$ 0.0041	0.1114 $\pm$ 0.0041	0.1177 $\pm$ 0.0047	0.1103 $\pm$ 0.0030	0.1103 $\pm$ 0.0030	0.1151 $\pm$ 0.0058
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.18 $\pm$ 1.27	89.00 $\pm$ 0.86	88.36 $\pm$ 1.04	88.53 $\pm$ 1.41	88.40 $\pm$ 0.55	88.52 $\pm$ 1.46	88.52 $\pm$ 1.31	88.88 $\pm$ 1.39	88.76 $\pm$ 1.28	88.29 $\pm$ 1.37
		AEOD $\downarrow$	0.0424 $\pm$ 0.0143	0.0431 $\pm$ 0.0131	0.0398 $\pm$ 0.0135	0.0337 $\pm$ 0.0137	0.0482 $\pm$ 0.0082	0.0452 $\pm$ 0.0150	0.0508 $\pm$ 0.0168	0.0367 $\pm$ 0.0061	0.0416 $\pm$ 0.0118	0.0564 $\pm$ 0.0186
		ASPD $\downarrow$	0.1053 $\pm$ 0.0090	0.1082 $\pm$ 0.0096	0.1015 $\pm$ 0.0091	0.0988 $\pm$ 0.0180	0.1108 $\pm$ 0.0096	0.1050 $\pm$ 0.0192	0.1108 $\pm$ 0.0130	0.1031 $\pm$ 0.0134	0.1035 $\pm$ 0.0113	0.1148 $\pm$ 0.0213
Robust FL	LASA*	ACC (%) $\uparrow$	89.52 $\pm$ 0.91	89.27 $\pm$ 0.69	85.62 $\pm$ 1.12	85.64 $\pm$ 1.70	87.87 $\pm$ 1.39	89.25 $\pm$ 0.66	89.61 $\pm$ 0.88	86.06 $\pm$ 1.23	86.04 $\pm$ 1.29	88.11 $\pm$ 1.33
		AEOD $\downarrow$	0.0374 $\pm$ 0.0060	0.0478 $\pm$ 0.0083	0.0257 $\pm$ 0.0069	0.0352 $\pm$ 0.0106	0.0449 $\pm$ 0.0092	0.0369 $\pm$ 0.0054	0.0476 $\pm$ 0.0076	0.0281 $\pm$ 0.0075	0.0307 $\pm$ 0.0020	0.0404 $\pm$ 0.0030
		ASPD $\downarrow$	0.1078 $\pm$ 0.0033	0.1158 $\pm$ 0.0048	0.0732 $\pm$ 0.0112	0.0811 $\pm$ 0.0151	0.1022 $\pm$ 0.0128	0.1048 $\pm$ 0.0087	0.1158 $\pm$ 0.0046	0.0751 $\pm$ 0.0137	0.0779 $\pm$ 0.0075	0.1011 $\pm$ 0.0082
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.01 $\pm$ 0.92	89.28 $\pm$ 1.48	49.97 $\pm$ 1.80	51.67 $\pm$ 0.00	82.74 $\pm$ 2.15	89.09 $\pm$ 1.11	89.78 $\pm$ 0.80	61.06 $\pm$ 17.25	63.21 $\pm$ 15.26	79.49 $\pm$ 15.09
		AEOD $\downarrow$	0.0308 $\pm$ 0.0028	0.0372 $\pm$ 0.0093	0.0069 $\pm$ 0.0170	0.0000 $\pm$ 0.0000	0.0183 $\pm$ 0.0090	0.0305 $\pm$ 0.0043	0.0353 $\pm$ 0.0094	0.0069 $\pm$ 0.0083	0.0144 $\pm$ 0.0191	0.0182 $\pm$ 0.0102
		ASPD $\downarrow$	0.0969 $\pm$ 0.0048	0.1062 $\pm$ 0.0111	0.0082 $\pm$ 0.0201	0.0000 $\pm$ 0.0000	0.0446 $\pm$ 0.0234	0.0996 $\pm$ 0.0103	0.1063 $\pm$ 0.0074	0.0170 $\pm$ 0.0214	0.0215 $\pm$ 0.0317	0.0594 $\pm$ 0.0181
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.99 $\pm$ 0.65	90.32 $\pm$ 0.71	79.32 $\pm$ 12.97	82.14 $\pm$ 14.30	88.87 $\pm$ 1.34	90.39 $\pm$ 0.39	90.12 $\pm$ 1.74	84.46 $\pm$ 5.99	84.73 $\pm$ 3.40	84.66 $\pm$ 9.74
		AEOD $\downarrow$	0.0359 $\pm$ 0.0077	0.0377 $\pm$ 0.0062	0.0223 $\pm$ 0.0112	0.0321 $\pm$ 0.0091	0.0422 $\pm$ 0.0103	0.0351 $\pm$ 0.0063	0.0417 $\pm$ 0.0069	0.0462 $\pm$ 0.0214	0.0320 $\pm$ 0.0093	0.0433 $\pm$ 0.0137
		ASPD $\downarrow$	0.1073 $\pm$ 0.0050	0.1132 $\pm$ 0.0096	0.0615 $\pm$ 0.0351	0.0800 $\pm$ 0.0345	0.1050 $\pm$ 0.0104	0.1122 $\pm$ 0.0054	0.1152 $\pm$ 0.0109	0.0966 $\pm$ 0.0220	0.0757 $\pm$ 0.0155	0.1013 $\pm$ 0.0128
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.81 $\pm$ 0.63	88.65 $\pm$ 0.88	88.89 $\pm$ 0.80	89.17 $\pm$ 0.78	88.71 $\pm$ 0.93	88.88 $\pm$ 1.42	88.74 $\pm$ 0.84	88.80 $\pm$ 0.82	88.73 $\pm$ 0.75	88.82 $\pm$ 1.07
		AEOD $\downarrow$	0.0423 $\pm$ 0.0127	0.0365 $\pm$ 0.0094	0.0345 $\pm$ 0.0052	0.0368 $\pm$ 0.0043	0.0345 $\pm$ 0.0138	0.0286 $\pm$ 0.0049	0.0286 $\pm$ 0.0080	0.0321 $\pm$ 0.0077	0.0352 $\pm$ 0.0066	0.0294 $\pm$ 0.0048
		ASPD $\downarrow$	0.1071 $\pm$ 0.0086	0.1011 $\pm$ 0.0069	0.1001 $\pm$ 0.0083	0.1037 $\pm$ 0.0056	0.0985 $\pm$ 0.0128	0.0967 $\pm$ 0.0085	0.0966 $\pm$ 0.0026	0.0966 $\pm$ 0.0066	0.0998 $\pm$ 0.0075	0.0971 $\pm$ 0.0094

All displayed cells use the same 6 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Raw view: all nine methods use uncalibrated argmax (margin > 0); same saved checkpoint per ID.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Native method rules

10 shared seeds: 91001–91010; round 70; validation split (19,867 images)

Table 4: CelebA [native]: IID / non-IID. COMPLETED VALIDATION COHORT – 10 shared seeds (91001–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	84.72 $\pm$ 11.74	89.57 $\pm$ 0.78	68.19 $\pm$ 5.77	66.01 $\pm$ 3.57	84.37 $\pm$ 2.38	86.85 $\pm$ 5.45	89.51 $\pm$ 1.82	74.76 $\pm$ 13.38	74.51 $\pm$ 13.37	85.48 $\pm$ 2.58
		AEOD $\downarrow$	0.0384 $\pm$ 0.0176	0.0519 $\pm$ 0.0096	0.1004 $\pm$ 0.0410	0.0950 $\pm$ 0.0477	0.0280 $\pm$ 0.0076	0.0428 $\pm$ 0.0137	0.0492 $\pm$ 0.0045	0.0644 $\pm$ 0.1097	0.0522 $\pm$ 0.0913	0.0267 $\pm$ 0.0157
		ASPD $\downarrow$	0.0937 $\pm$ 0.0366	0.1212 $\pm$ 0.0064	0.0993 $\pm$ 0.0380	0.0807 $\pm$ 0.0413	0.0693 $\pm$ 0.0161	0.0958 $\pm$ 0.0353	0.1161 $\pm$ 0.0143	0.0614 $\pm$ 0.0893	0.0591 $\pm$ 0.0756	0.0741 $\pm$ 0.0247
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.47 $\pm$ 0.85	89.79 $\pm$ 0.76	69.47 $\pm$ 4.65	62.59 $\pm$ 5.11	85.55 $\pm$ 2.31	89.59 $\pm$ 0.97	89.83 $\pm$ 0.92	71.45 $\pm$ 11.85	67.06 $\pm$ 11.91	86.18 $\pm$ 2.41
		AEOD $\downarrow$	0.0414 $\pm$ 0.0076	0.0489 $\pm$ 0.0042	0.1158 $\pm$ 0.0635	0.1594 $\pm$ 0.0822	0.0288 $\pm$ 0.0081	0.0388 $\pm$ 0.0057	0.0462 $\pm$ 0.0081	0.0759 $\pm$ 0.0860	0.1307 $\pm$ 0.1184	0.0328 $\pm$ 0.0133
		ASPD $\downarrow$	0.1102 $\pm$ 0.0077	0.1188 $\pm$ 0.0043	0.1120 $\pm$ 0.0554	0.1461 $\pm$ 0.0668	0.0781 $\pm$ 0.0116	0.1092 $\pm$ 0.0071	0.1163 $\pm$ 0.0071	0.0730 $\pm$ 0.0885	0.1171 $\pm$ 0.1067	0.0829 $\pm$ 0.0204
Robust FL	Median	ACC (%) $\uparrow$	86.28 $\pm$ 1.31	86.52 $\pm$ 1.35	84.66 $\pm$ 1.47	85.00 $\pm$ 1.38	85.54 $\pm$ 1.50	86.22 $\pm$ 1.31	86.51 $\pm$ 1.24	84.47 $\pm$ 1.30	84.48 $\pm$ 1.35	85.47 $\pm$ 1.26
		AEOD $\downarrow$	0.0510 $\pm$ 0.0081	0.0610 $\pm$ 0.0087	0.0438 $\pm$ 0.0135	0.0484 $\pm$ 0.0090	0.0534 $\pm$ 0.0096	0.0518 $\pm$ 0.0068	0.0594 $\pm$ 0.0084	0.0425 $\pm$ 0.0118	0.0461 $\pm$ 0.0142	0.0502 $\pm$ 0.0111
		ASPD $\downarrow$	0.0980 $\pm$ 0.0117	0.1062 $\pm$ 0.0111	0.0825 $\pm$ 0.0142	0.0852 $\pm$ 0.0127	0.0965 $\pm$ 0.0124	0.0963 $\pm$ 0.0096	0.1067 $\pm$ 0.0097	0.0799 $\pm$ 0.0113	0.0839 $\pm$ 0.0124	0.0929 $\pm$ 0.0086
Robust FL	FLTrust	ACC (%) $\uparrow$	89.43 $\pm$ 0.75	89.67 $\pm$ 0.82	89.68 $\pm$ 0.74	89.68 $\pm$ 0.74	89.72 $\pm$ 0.64	89.68 $\pm$ 0.60	89.83 $\pm$ 0.69	89.48 $\pm$ 0.70	89.48 $\pm$ 0.70	89.64 $\pm$ 0.71
		AEOD $\downarrow$	0.0432 $\pm$ 0.0075	0.0518 $\pm$ 0.0072	0.0422 $\pm$ 0.0069	0.0422 $\pm$ 0.0069	0.0496 $\pm$ 0.0061	0.0445 $\pm$ 0.0049	0.0493 $\pm$ 0.0054	0.0441 $\pm$ 0.0063	0.0441 $\pm$ 0.0063	0.0475 $\pm$ 0.0052
		ASPD $\downarrow$	0.1111 $\pm$ 0.0062	0.1207 $\pm$ 0.0064	0.1119 $\pm$ 0.0071	0.1119 $\pm$ 0.0071	0.1177 $\pm$ 0.0057	0.1132 $\pm$ 0.0060	0.1197 $\pm$ 0.0044	0.1120 $\pm$ 0.0041	0.1120 $\pm$ 0.0041	0.1153 $\pm$ 0.0053
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.35 $\pm$ 1.33	89.06 $\pm$ 0.92	88.80 $\pm$ 1.06	88.82 $\pm$ 1.21	88.60 $\pm$ 1.12	88.60 $\pm$ 1.14	88.56 $\pm$ 1.19	88.53 $\pm$ 1.31	88.52 $\pm$ 1.60	88.39 $\pm$ 1.14
		AEOD $\downarrow$	0.0455 $\pm$ 0.0116	0.0442 $\pm$ 0.0137	0.0423 $\pm$ 0.0116	0.0385 $\pm$ 0.0131	0.0493 $\pm$ 0.0089	0.0456 $\pm$ 0.0125	0.0491 $\pm$ 0.0132	0.0421 $\pm$ 0.0231	0.0409 $\pm$ 0.0122	0.0597 $\pm$ 0.0189
		ASPD $\downarrow$	0.1083 $\pm$ 0.0100	0.1097 $\pm$ 0.0143	0.1071 $\pm$ 0.0115	0.1042 $\pm$ 0.0152	0.1136 $\pm$ 0.0095	0.1069 $\pm$ 0.0151	0.1110 $\pm$ 0.0103	0.1031 $\pm$ 0.0212	0.1020 $\pm$ 0.0149	0.1188 $\pm$ 0.0190
Robust FL	LASA*	ACC (%) $\uparrow$	89.39 $\pm$ 1.42	89.46 $\pm$ 0.79	85.75 $\pm$ 1.71	85.96 $\pm$ 1.62	88.20 $\pm$ 1.40	89.27 $\pm$ 0.85	89.51 $\pm$ 0.88	85.91 $\pm$ 1.69	85.93 $\pm$ 1.64	88.15 $\pm$ 1.22
		AEOD $\downarrow$	0.0377 $\pm$ 0.0049	0.0493 $\pm$ 0.0066	0.0330 $\pm$ 0.0109	0.0366 $\pm$ 0.0105	0.0450 $\pm$ 0.0073	0.0386 $\pm$ 0.0057	0.0500 $\pm$ 0.0096	0.0301 $\pm$ 0.0069	0.0345 $\pm$ 0.0113	0.0417 $\pm$ 0.0037
		ASPD $\downarrow$	0.1074 $\pm$ 0.0088	0.1176 $\pm$ 0.0059	0.0811 $\pm$ 0.0138	0.0839 $\pm$ 0.0152	0.1050 $\pm$ 0.0113	0.1070 $\pm$ 0.0084	0.1181 $\pm$ 0.0059	0.0767 $\pm$ 0.0129	0.0791 $\pm$ 0.0128	0.1026 $\pm$ 0.0074
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.18 $\pm$ 0.83	89.57 $\pm$ 1.24	49.32 $\pm$ 1.59	50.33 $\pm$ 1.72	83.20 $\pm$ 1.76	89.39 $\pm$ 1.11	89.84 $\pm$ 0.76	59.95 $\pm$ 16.51	61.29 $\pm$ 15.59	78.37 $\pm$ 14.94
		AEOD $\downarrow$	0.0317 $\pm$ 0.0026	0.0369 $\pm$ 0.0071	0.0042 $\pm$ 0.0132	0.0000 $\pm$ 0.0000	0.0205 $\pm$ 0.0098	0.0314 $\pm$ 0.0054	0.0394 $\pm$ 0.0103	0.0066 $\pm$ 0.0094	0.0108 $\pm$ 0.0161	0.0209 $\pm$ 0.0104
		ASPD $\downarrow$	0.0998 $\pm$ 0.0062	0.1078 $\pm$ 0.0092	0.0049 $\pm$ 0.0156	0.0000 $\pm$ 0.0000	0.0491 $\pm$ 0.0199	0.1026 $\pm$ 0.0086	0.1109 $\pm$ 0.0090	0.0169 $\pm$ 0.0251	0.0206 $\pm$ 0.0317	0.0552 $\pm$ 0.0238
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.93 $\pm$ 1.10	90.37 $\pm$ 0.60	77.83 $\pm$ 12.15	75.99 $\pm$ 16.52	86.44 $\pm$ 5.76	89.69 $\pm$ 1.68	90.07 $\pm$ 1.34	81.90 $\pm$ 11.66	80.44 $\pm$ 12.06	79.69 $\pm$ 13.94
		AEOD $\downarrow$	0.0365 $\pm$ 0.0075	0.0363 $\pm$ 0.0086	0.0298 $\pm$ 0.0299	0.0293 $\pm$ 0.0146	0.0368 $\pm$ 0.0144	0.0357 $\pm$ 0.0055	0.0426 $\pm$ 0.0062	0.0400 $\pm$ 0.0214	0.0359 $\pm$ 0.0262	0.0408 $\pm$ 0.0124
		ASPD $\downarrow$	0.1074 $\pm$ 0.0113	0.1118 $\pm$ 0.0100	0.0607 $\pm$ 0.0354	0.0652 $\pm$ 0.0419	0.0903 $\pm$ 0.0325	0.1085 $\pm$ 0.0102	0.1144 $\pm$ 0.0088	0.0851 $\pm$ 0.0350	0.0763 $\pm$ 0.0174	0.0892 $\pm$ 0.0276
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.26 $\pm$ 1.04	88.39 $\pm$ 0.63	88.47 $\pm$ 0.91	88.69 $\pm$ 0.81	88.38 $\pm$ 0.89	88.59 $\pm$ 1.17	88.44 $\pm$ 0.74	88.41 $\pm$ 0.92	88.22 $\pm$ 0.78	88.34 $\pm$ 0.96
		AEOD $\downarrow$	0.0097 $\pm$ 0.0075	0.0107 $\pm$ 0.0096	0.0061 $\pm$ 0.0042	0.0077 $\pm$ 0.0046	0.0165 $\pm$ 0.0072	0.0070 $\pm$ 0.0042	0.0100 $\pm$ 0.0070	0.0078 $\pm$ 0.0056	0.0126 $\pm$ 0.0093	0.0087 $\pm$ 0.0066
		ASPD $\downarrow$	0.0625 $\pm$ 0.0138	0.0607 $\pm$ 0.0103	0.0640 $\pm$ 0.0142	0.0638 $\pm$ 0.0070	0.0559 $\pm$ 0.0156	0.0651 $\pm$ 0.0090	0.0589 $\pm$ 0.0092	0.0616 $\pm$ 0.0141	0.0584 $\pm$ 0.0189	0.0596 $\pm$ 0.0146

All displayed cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
This table uses 886 cu128 and 14 cu130 records; the source cohort contains 14 cu130 records. No 70-round equivalence claim.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Native view: GuardFed uses saved clean-training-root group calibration; the eight baselines use native argmax.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Native method rules

9 shared seeds: 91002–91010; round 70; validation split (19,867 images)

Table 5: CelebA [native]: IID / non-IID. COMPLETED VALIDATION COHORT – 9 shared seeds (91002–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	88.39 $\pm$ 1.84	89.72 $\pm$ 0.65	69.08 $\pm$ 5.35	66.61 $\pm$ 3.21	84.91 $\pm$ 1.77	88.09 $\pm$ 4.01	89.91 $\pm$ 1.39	75.58 $\pm$ 13.92	75.31 $\pm$ 13.93	85.69 $\pm$ 2.65
		AEOD $\downarrow$	0.0426 $\pm$ 0.0119	0.0522 $\pm$ 0.0101	0.0939 $\pm$ 0.0376	0.0904 $\pm$ 0.0481	0.0289 $\pm$ 0.0075	0.0457 $\pm$ 0.0107	0.0485 $\pm$ 0.0043	0.0560 $\pm$ 0.1128	0.0510 $\pm$ 0.0968	0.0273 $\pm$ 0.0165
		ASPD $\downarrow$	0.1041 $\pm$ 0.0171	0.1225 $\pm$ 0.0053	0.0962 $\pm$ 0.0389	0.0781 $\pm$ 0.0429	0.0702 $\pm$ 0.0168	0.1050 $\pm$ 0.0215	0.1190 $\pm$ 0.0117	0.0552 $\pm$ 0.0923	0.0603 $\pm$ 0.0801	0.0748 $\pm$ 0.0260
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.61 $\pm$ 0.78	89.91 $\pm$ 0.71	70.47 $\pm$ 3.62	63.49 $\pm$ 4.50	86.17 $\pm$ 1.29	89.76 $\pm$ 0.83	89.92 $\pm$ 0.92	72.23 $\pm$ 12.29	67.94 $\pm$ 12.28	86.50 $\pm$ 2.31
		AEOD $\downarrow$	0.0411 $\pm$ 0.0080	0.0483 $\pm$ 0.0040	0.1030 $\pm$ 0.0521	0.1435 $\pm$ 0.0691	0.0301 $\pm$ 0.0076	0.0380 $\pm$ 0.0054	0.0448 $\pm$ 0.0073	0.0688 $\pm$ 0.0881	0.1327 $\pm$ 0.1254	0.0318 $\pm$ 0.0137
		ASPD $\downarrow$	0.1108 $\pm$ 0.0078	0.1189 $\pm$ 0.0046	0.1009 $\pm$ 0.0454	0.1346 $\pm$ 0.0595	0.0797 $\pm$ 0.0111	0.1095 $\pm$ 0.0074	0.1158 $\pm$ 0.0073	0.0676 $\pm$ 0.0922	0.1180 $\pm$ 0.1131	0.0838 $\pm$ 0.0214
Robust FL	Median	ACC (%) $\uparrow$	86.51 $\pm$ 1.16	86.71 $\pm$ 1.30	84.87 $\pm$ 1.39	85.25 $\pm$ 1.20	85.73 $\pm$ 1.47	86.40 $\pm$ 1.26	86.51 $\pm$ 1.31	84.54 $\pm$ 1.36	84.59 $\pm$ 1.39	85.54 $\pm$ 1.32
		AEOD $\downarrow$	0.0509 $\pm$ 0.0086	0.0610 $\pm$ 0.0093	0.0433 $\pm$ 0.0142	0.0483 $\pm$ 0.0096	0.0532 $\pm$ 0.0101	0.0510 $\pm$ 0.0066	0.0607 $\pm$ 0.0078	0.0425 $\pm$ 0.0125	0.0448 $\pm$ 0.0144	0.0503 $\pm$ 0.0118
		ASPD $\downarrow$	0.0988 $\pm$ 0.0121	0.1070 $\pm$ 0.0114	0.0830 $\pm$ 0.0150	0.0860 $\pm$ 0.0132	0.0974 $\pm$ 0.0128	0.0965 $\pm$ 0.0102	0.1074 $\pm$ 0.0100	0.0800 $\pm$ 0.0120	0.0833 $\pm$ 0.0130	0.0930 $\pm$ 0.0091
Robust FL	FLTrust	ACC (%) $\uparrow$	89.47 $\pm$ 0.78	89.73 $\pm$ 0.85	89.73 $\pm$ 0.76	89.73 $\pm$ 0.76	89.79 $\pm$ 0.64	89.77 $\pm$ 0.56	89.87 $\pm$ 0.72	89.55 $\pm$ 0.70	89.55 $\pm$ 0.70	89.68 $\pm$ 0.74
		AEOD $\downarrow$	0.0420 $\pm$ 0.0069	0.0521 $\pm$ 0.0075	0.0416 $\pm$ 0.0071	0.0416 $\pm$ 0.0071	0.0494 $\pm$ 0.0064	0.0450 $\pm$ 0.0049	0.0488 $\pm$ 0.0056	0.0441 $\pm$ 0.0066	0.0441 $\pm$ 0.0066	0.0481 $\pm$ 0.0051
		ASPD $\downarrow$	0.1108 $\pm$ 0.0065	0.1214 $\pm$ 0.0064	0.1124 $\pm$ 0.0074	0.1124 $\pm$ 0.0074	0.1181 $\pm$ 0.0059	0.1142 $\pm$ 0.0055	0.1194 $\pm$ 0.0046	0.1124 $\pm$ 0.0041	0.1124 $\pm$ 0.0041	0.1161 $\pm$ 0.0049
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.58 $\pm$ 1.18	89.24 $\pm$ 0.77	88.85 $\pm$ 1.11	88.90 $\pm$ 1.26	88.86 $\pm$ 0.81	88.67 $\pm$ 1.19	88.73 $\pm$ 1.13	88.74 $\pm$ 1.21	88.89 $\pm$ 1.17	88.46 $\pm$ 1.18
		AEOD $\downarrow$	0.0446 $\pm$ 0.0120	0.0455 $\pm$ 0.0139	0.0428 $\pm$ 0.0122	0.0363 $\pm$ 0.0116	0.0478 $\pm$ 0.0078	0.0464 $\pm$ 0.0130	0.0490 $\pm$ 0.0140	0.0451 $\pm$ 0.0223	0.0421 $\pm$ 0.0123	0.0606 $\pm$ 0.0198
		ASPD $\downarrow$	0.1093 $\pm$ 0.0101	0.1124 $\pm$ 0.0121	0.1076 $\pm$ 0.0121	0.1032 $\pm$ 0.0158	0.1136 $\pm$ 0.0101	0.1072 $\pm$ 0.0159	0.1117 $\pm$ 0.0106	0.1077 $\pm$ 0.0166	0.1056 $\pm$ 0.0100	0.1198 $\pm$ 0.0199
Robust FL	LASA*	ACC (%) $\uparrow$	89.77 $\pm$ 0.81	89.59 $\pm$ 0.72	86.15 $\pm$ 1.21	86.15 $\pm$ 1.61	88.38 $\pm$ 1.35	89.46 $\pm$ 0.63	89.68 $\pm$ 0.72	86.33 $\pm$ 1.10	86.32 $\pm$ 1.16	88.34 $\pm$ 1.13
		AEOD $\downarrow$	0.0373 $\pm$ 0.0050	0.0489 $\pm$ 0.0069	0.0315 $\pm$ 0.0103	0.0379 $\pm$ 0.0103	0.0446 $\pm$ 0.0077	0.0379 $\pm$ 0.0056	0.0485 $\pm$ 0.0090	0.0301 $\pm$ 0.0074	0.0350 $\pm$ 0.0118	0.0409 $\pm$ 0.0028
		ASPD $\downarrow$	0.1099 $\pm$ 0.0042	0.1184 $\pm$ 0.0057	0.0810 $\pm$ 0.0146	0.0861 $\pm$ 0.0143	0.1059 $\pm$ 0.0116	0.1078 $\pm$ 0.0085	0.1179 $\pm$ 0.0062	0.0782 $\pm$ 0.0127	0.0818 $\pm$ 0.0101	0.1033 $\pm$ 0.0075
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.27 $\pm$ 0.84	89.65 $\pm$ 1.29	49.42 $\pm$ 1.64	50.56 $\pm$ 1.67	83.20 $\pm$ 1.87	89.52 $\pm$ 1.09	89.95 $\pm$ 0.71	61.24 $\pm$ 16.96	62.72 $\pm$ 15.82	77.66 $\pm$ 15.66
		AEOD $\downarrow$	0.0316 $\pm$ 0.0028	0.0369 $\pm$ 0.0075	0.0046 $\pm$ 0.0139	0.0000 $\pm$ 0.0000	0.0185 $\pm$ 0.0080	0.0301 $\pm$ 0.0039	0.0380 $\pm$ 0.0099	0.0073 $\pm$ 0.0097	0.0119 $\pm$ 0.0166	0.0188 $\pm$ 0.0084
		ASPD $\downarrow$	0.1002 $\pm$ 0.0064	0.1085 $\pm$ 0.0094	0.0055 $\pm$ 0.0164	0.0000 $\pm$ 0.0000	0.0471 $\pm$ 0.0199	0.1023 $\pm$ 0.0091	0.1104 $\pm$ 0.0094	0.0188 $\pm$ 0.0259	0.0226 $\pm$ 0.0329	0.0523 $\pm$ 0.0233
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	90.22 $\pm$ 0.65	90.44 $\pm$ 0.59	76.75 $\pm$ 12.37	74.59 $\pm$ 16.88	86.38 $\pm$ 6.11	90.20 $\pm$ 0.46	90.17 $\pm$ 1.38	81.41 $\pm$ 12.25	79.86 $\pm$ 12.64	78.76 $\pm$ 14.45
		AEOD $\downarrow$	0.0371 $\pm$ 0.0076	0.0362 $\pm$ 0.0091	0.0308 $\pm$ 0.0315	0.0298 $\pm$ 0.0154	0.0367 $\pm$ 0.0153	0.0359 $\pm$ 0.0058	0.0415 $\pm$ 0.0055	0.0407 $\pm$ 0.0225	0.0371 $\pm$ 0.0275	0.0404 $\pm$ 0.0131
		ASPD $\downarrow$	0.1101 $\pm$ 0.0078	0.1131 $\pm$ 0.0098	0.0588 $\pm$ 0.0369	0.0627 $\pm$ 0.0437	0.0903 $\pm$ 0.0345	0.1114 $\pm$ 0.0045	0.1154 $\pm$ 0.0087	0.0859 $\pm$ 0.0370	0.0780 $\pm$ 0.0177	0.0882 $\pm$ 0.0291
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.35 $\pm$ 1.06	88.41 $\pm$ 0.66	88.62 $\pm$ 0.82	88.84 $\pm$ 0.68	88.55 $\pm$ 0.77	88.59 $\pm$ 1.24	88.50 $\pm$ 0.76	88.39 $\pm$ 0.97	88.24 $\pm$ 0.82	88.47 $\pm$ 0.94
		AEOD $\downarrow$	0.0097 $\pm$ 0.0080	0.0115 $\pm$ 0.0098	0.0054 $\pm$ 0.0039	0.0068 $\pm$ 0.0038	0.0171 $\pm$ 0.0074	0.0063 $\pm$ 0.0039	0.0104 $\pm$ 0.0073	0.0083 $\pm$ 0.0057	0.0119 $\pm$ 0.0095	0.0079 $\pm$ 0.0065
		ASPD $\downarrow$	0.0623 $\pm$ 0.0146	0.0604 $\pm$ 0.0108	0.0667 $\pm$ 0.0120	0.0633 $\pm$ 0.0072	0.0555 $\pm$ 0.0165	0.0638 $\pm$ 0.0086	0.0579 $\pm$ 0.0090	0.0609 $\pm$ 0.0148	0.0605 $\pm$ 0.0188	0.0613 $\pm$ 0.0144

All displayed cells use the same 9 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Native view: GuardFed uses saved clean-training-root group calibration; the eight baselines use native argmax.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Native method rules

6 shared seeds: 91005–91010; round 70; validation split (19,867 images)

Table 6: CelebA [native]: IID / non-IID. COMPLETED VALIDATION COHORT – 6 shared seeds (91005–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	87.99 $\pm$ 2.11	89.71 $\pm$ 0.80	69.36 $\pm$ 6.14	67.41 $\pm$ 3.17	85.70 $\pm$ 1.38	86.90 $\pm$ 4.54	89.58 $\pm$ 1.62	78.23 $\pm$ 12.19	78.69 $\pm$ 12.03	86.13 $\pm$ 3.06
		AEOD $\downarrow$	0.0406 $\pm$ 0.0131	0.0535 $\pm$ 0.0119	0.0852 $\pm$ 0.0377	0.0823 $\pm$ 0.0434	0.0280 $\pm$ 0.0074	0.0464 $\pm$ 0.0125	0.0479 $\pm$ 0.0036	0.0756 $\pm$ 0.1376	0.0652 $\pm$ 0.1170	0.0312 $\pm$ 0.0166
		ASPD $\downarrow$	0.0992 $\pm$ 0.0184	0.1227 $\pm$ 0.0063	0.0866 $\pm$ 0.0413	0.0698 $\pm$ 0.0304	0.0738 $\pm$ 0.0142	0.0977 $\pm$ 0.0234	0.1151 $\pm$ 0.0128	0.0788 $\pm$ 0.1078	0.0804 $\pm$ 0.0927	0.0817 $\pm$ 0.0283
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.39 $\pm$ 0.89	89.65 $\pm$ 0.75	70.49 $\pm$ 3.83	65.00 $\pm$ 4.85	85.87 $\pm$ 1.25	89.56 $\pm$ 0.96	89.73 $\pm$ 1.11	75.25 $\pm$ 10.47	70.49 $\pm$ 12.37	86.17 $\pm$ 2.80
		AEOD $\downarrow$	0.0414 $\pm$ 0.0096	0.0480 $\pm$ 0.0047	0.0912 $\pm$ 0.0604	0.1179 $\pm$ 0.0658	0.0305 $\pm$ 0.0060	0.0388 $\pm$ 0.0033	0.0431 $\pm$ 0.0055	0.0727 $\pm$ 0.1009	0.0952 $\pm$ 0.1115	0.0299 $\pm$ 0.0162
		ASPD $\downarrow$	0.1090 $\pm$ 0.0082	0.1170 $\pm$ 0.0044	0.0961 $\pm$ 0.0536	0.1133 $\pm$ 0.0622	0.0790 $\pm$ 0.0076	0.1082 $\pm$ 0.0085	0.1133 $\pm$ 0.0064	0.0770 $\pm$ 0.1069	0.0879 $\pm$ 0.1083	0.0814 $\pm$ 0.0256
Robust FL	Median	ACC (%) $\uparrow$	86.37 $\pm$ 1.01	86.58 $\pm$ 1.26	84.48 $\pm$ 0.83	84.83 $\pm$ 0.80	85.45 $\pm$ 1.23	86.19 $\pm$ 1.06	86.20 $\pm$ 1.30	84.49 $\pm$ 1.15	84.57 $\pm$ 1.19	85.68 $\pm$ 1.19
		AEOD $\downarrow$	0.0504 $\pm$ 0.0088	0.0602 $\pm$ 0.0102	0.0402 $\pm$ 0.0154	0.0440 $\pm$ 0.0083	0.0526 $\pm$ 0.0099	0.0525 $\pm$ 0.0042	0.0579 $\pm$ 0.0076	0.0423 $\pm$ 0.0100	0.0433 $\pm$ 0.0112	0.0508 $\pm$ 0.0143
		ASPD $\downarrow$	0.0977 $\pm$ 0.0107	0.1054 $\pm$ 0.0109	0.0786 $\pm$ 0.0124	0.0807 $\pm$ 0.0088	0.0956 $\pm$ 0.0101	0.0969 $\pm$ 0.0033	0.1044 $\pm$ 0.0095	0.0786 $\pm$ 0.0089	0.0817 $\pm$ 0.0094	0.0950 $\pm$ 0.0072
Robust FL	FLTrust	ACC (%) $\uparrow$	89.41 $\pm$ 0.86	89.73 $\pm$ 0.92	89.61 $\pm$ 0.88	89.61 $\pm$ 0.88	89.67 $\pm$ 0.71	89.60 $\pm$ 0.61	89.75 $\pm$ 0.87	89.48 $\pm$ 0.63	89.48 $\pm$ 0.63	89.66 $\pm$ 0.82
		AEOD $\downarrow$	0.0392 $\pm$ 0.0056	0.0488 $\pm$ 0.0070	0.0385 $\pm$ 0.0059	0.0385 $\pm$ 0.0059	0.0475 $\pm$ 0.0063	0.0428 $\pm$ 0.0041	0.0482 $\pm$ 0.0064	0.0435 $\pm$ 0.0048	0.0435 $\pm$ 0.0048	0.0473 $\pm$ 0.0034
		ASPD $\downarrow$	0.1087 $\pm$ 0.0063	0.1189 $\pm$ 0.0046	0.1088 $\pm$ 0.0048	0.1088 $\pm$ 0.0048	0.1151 $\pm$ 0.0041	0.1114 $\pm$ 0.0041	0.1177 $\pm$ 0.0047	0.1103 $\pm$ 0.0030	0.1103 $\pm$ 0.0030	0.1151 $\pm$ 0.0058
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.18 $\pm$ 1.27	89.00 $\pm$ 0.86	88.36 $\pm$ 1.04	88.53 $\pm$ 1.41	88.40 $\pm$ 0.55	88.52 $\pm$ 1.46	88.52 $\pm$ 1.31	88.88 $\pm$ 1.39	88.76 $\pm$ 1.28	88.29 $\pm$ 1.37
		AEOD $\downarrow$	0.0424 $\pm$ 0.0143	0.0431 $\pm$ 0.0131	0.0398 $\pm$ 0.0135	0.0337 $\pm$ 0.0137	0.0482 $\pm$ 0.0082	0.0452 $\pm$ 0.0150	0.0508 $\pm$ 0.0168	0.0367 $\pm$ 0.0061	0.0416 $\pm$ 0.0118	0.0564 $\pm$ 0.0186
		ASPD $\downarrow$	0.1053 $\pm$ 0.0090	0.1082 $\pm$ 0.0096	0.1015 $\pm$ 0.0091	0.0988 $\pm$ 0.0180	0.1108 $\pm$ 0.0096	0.1050 $\pm$ 0.0192	0.1108 $\pm$ 0.0130	0.1031 $\pm$ 0.0134	0.1035 $\pm$ 0.0113	0.1148 $\pm$ 0.0213
Robust FL	LASA*	ACC (%) $\uparrow$	89.52 $\pm$ 0.91	89.27 $\pm$ 0.69	85.62 $\pm$ 1.12	85.64 $\pm$ 1.70	87.87 $\pm$ 1.39	89.25 $\pm$ 0.66	89.61 $\pm$ 0.88	86.06 $\pm$ 1.23	86.04 $\pm$ 1.29	88.11 $\pm$ 1.33
		AEOD $\downarrow$	0.0374 $\pm$ 0.0060	0.0478 $\pm$ 0.0083	0.0257 $\pm$ 0.0069	0.0352 $\pm$ 0.0106	0.0449 $\pm$ 0.0092	0.0369 $\pm$ 0.0054	0.0476 $\pm$ 0.0076	0.0281 $\pm$ 0.0075	0.0307 $\pm$ 0.0020	0.0404 $\pm$ 0.0030
		ASPD $\downarrow$	0.1078 $\pm$ 0.0033	0.1158 $\pm$ 0.0048	0.0732 $\pm$ 0.0112	0.0811 $\pm$ 0.0151	0.1022 $\pm$ 0.0128	0.1048 $\pm$ 0.0087	0.1158 $\pm$ 0.0046	0.0751 $\pm$ 0.0137	0.0779 $\pm$ 0.0075	0.1011 $\pm$ 0.0082
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.01 $\pm$ 0.92	89.28 $\pm$ 1.48	49.97 $\pm$ 1.80	51.67 $\pm$ 0.00	82.74 $\pm$ 2.15	89.09 $\pm$ 1.11	89.78 $\pm$ 0.80	61.06 $\pm$ 17.25	63.21 $\pm$ 15.26	79.49 $\pm$ 15.09
		AEOD $\downarrow$	0.0308 $\pm$ 0.0028	0.0372 $\pm$ 0.0093	0.0069 $\pm$ 0.0170	0.0000 $\pm$ 0.0000	0.0183 $\pm$ 0.0090	0.0305 $\pm$ 0.0043	0.0353 $\pm$ 0.0094	0.0069 $\pm$ 0.0083	0.0144 $\pm$ 0.0191	0.0182 $\pm$ 0.0102
		ASPD $\downarrow$	0.0969 $\pm$ 0.0048	0.1062 $\pm$ 0.0111	0.0082 $\pm$ 0.0201	0.0000 $\pm$ 0.0000	0.0446 $\pm$ 0.0234	0.0996 $\pm$ 0.0103	0.1063 $\pm$ 0.0074	0.0170 $\pm$ 0.0214	0.0215 $\pm$ 0.0317	0.0594 $\pm$ 0.0181
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.99 $\pm$ 0.65	90.32 $\pm$ 0.71	79.32 $\pm$ 12.97	82.14 $\pm$ 14.30	88.87 $\pm$ 1.34	90.39 $\pm$ 0.39	90.12 $\pm$ 1.74	84.46 $\pm$ 5.99	84.73 $\pm$ 3.40	84.66 $\pm$ 9.74
		AEOD $\downarrow$	0.0359 $\pm$ 0.0077	0.0377 $\pm$ 0.0062	0.0223 $\pm$ 0.0112	0.0321 $\pm$ 0.0091	0.0422 $\pm$ 0.0103	0.0351 $\pm$ 0.0063	0.0417 $\pm$ 0.0069	0.0462 $\pm$ 0.0214	0.0320 $\pm$ 0.0093	0.0433 $\pm$ 0.0137
		ASPD $\downarrow$	0.1073 $\pm$ 0.0050	0.1132 $\pm$ 0.0096	0.0615 $\pm$ 0.0351	0.0800 $\pm$ 0.0345	0.1050 $\pm$ 0.0104	0.1122 $\pm$ 0.0054	0.1152 $\pm$ 0.0109	0.0966 $\pm$ 0.0220	0.0757 $\pm$ 0.0155	0.1013 $\pm$ 0.0128
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.56 $\pm$ 0.68	88.29 $\pm$ 0.75	88.57 $\pm$ 0.87	88.75 $\pm$ 0.79	88.35 $\pm$ 0.87	88.58 $\pm$ 1.42	88.43 $\pm$ 0.87	88.49 $\pm$ 0.88	88.42 $\pm$ 0.83	88.42 $\pm$ 1.10
		AEOD $\downarrow$	0.0114 $\pm$ 0.0073	0.0148 $\pm$ 0.0104	0.0055 $\pm$ 0.0036	0.0070 $\pm$ 0.0040	0.0171 $\pm$ 0.0083	0.0062 $\pm$ 0.0039	0.0110 $\pm$ 0.0086	0.0083 $\pm$ 0.0044	0.0067 $\pm$ 0.0068	0.0089 $\pm$ 0.0080
		ASPD $\downarrow$	0.0665 $\pm$ 0.0159	0.0581 $\pm$ 0.0129	0.0697 $\pm$ 0.0110	0.0617 $\pm$ 0.0085	0.0559 $\pm$ 0.0203	0.0641 $\pm$ 0.0067	0.0570 $\pm$ 0.0090	0.0566 $\pm$ 0.0081	0.0611 $\pm$ 0.0114	0.0583 $\pm$ 0.0172

All displayed cells use the same 6 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Native view: GuardFed uses saved clean-training-root group calibration; the eight baselines use native argmax.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Shared calibration

10 shared seeds: 91001–91010; round 70; validation split (19,867 images)

Table 7: CelebA [shared calibration]: IID / non-IID. COMPLETED VALIDATION COHORT – 10 shared seeds (91001–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	84.38 $\pm$ 11.63	89.17 $\pm$ 0.91	68.98 $\pm$ 5.78	66.69 $\pm$ 3.47	84.04 $\pm$ 2.31	86.50 $\pm$ 5.47	89.07 $\pm$ 1.89	75.03 $\pm$ 12.51	74.59 $\pm$ 12.78	85.01 $\pm$ 2.51
		AEOD $\downarrow$	0.0061 $\pm$ 0.0058	0.0104 $\pm$ 0.0075	0.0114 $\pm$ 0.0079	0.0099 $\pm$ 0.0083	0.0115 $\pm$ 0.0061	0.0080 $\pm$ 0.0052	0.0080 $\pm$ 0.0066	0.0101 $\pm$ 0.0068	0.0110 $\pm$ 0.0073	0.0161 $\pm$ 0.0112
		ASPD $\downarrow$	0.0546 $\pm$ 0.0251	0.0689 $\pm$ 0.0156	0.0085 $\pm$ 0.0070	0.0141 $\pm$ 0.0121	0.0299 $\pm$ 0.0173	0.0586 $\pm$ 0.0248	0.0689 $\pm$ 0.0175	0.0127 $\pm$ 0.0109	0.0168 $\pm$ 0.0122	0.0333 $\pm$ 0.0219
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.11 $\pm$ 0.83	89.36 $\pm$ 0.77	69.73 $\pm$ 4.94	63.30 $\pm$ 4.77	85.17 $\pm$ 2.24	89.22 $\pm$ 1.04	89.55 $\pm$ 0.96	71.78 $\pm$ 11.30	67.66 $\pm$ 11.30	85.84 $\pm$ 2.32
		AEOD $\downarrow$	0.0095 $\pm$ 0.0067	0.0076 $\pm$ 0.0074	0.0091 $\pm$ 0.0068	0.0110 $\pm$ 0.0099	0.0088 $\pm$ 0.0080	0.0088 $\pm$ 0.0049	0.0087 $\pm$ 0.0065	0.0090 $\pm$ 0.0073	0.0085 $\pm$ 0.0066	0.0102 $\pm$ 0.0035
		ASPD $\downarrow$	0.0704 $\pm$ 0.0089	0.0752 $\pm$ 0.0054	0.0071 $\pm$ 0.0061	0.0124 $\pm$ 0.0066	0.0417 $\pm$ 0.0112	0.0712 $\pm$ 0.0083	0.0760 $\pm$ 0.0084	0.0054 $\pm$ 0.0049	0.0087 $\pm$ 0.0071	0.0534 $\pm$ 0.0169
Robust FL	Median	ACC (%) $\uparrow$	85.87 $\pm$ 1.33	86.20 $\pm$ 1.40	84.30 $\pm$ 1.57	84.56 $\pm$ 1.36	85.20 $\pm$ 1.52	85.84 $\pm$ 1.25	86.11 $\pm$ 1.25	84.12 $\pm$ 1.25	84.15 $\pm$ 1.35	85.08 $\pm$ 1.28
		AEOD $\downarrow$	0.0118 $\pm$ 0.0067	0.0132 $\pm$ 0.0078	0.0120 $\pm$ 0.0125	0.0082 $\pm$ 0.0080	0.0130 $\pm$ 0.0074	0.0108 $\pm$ 0.0071	0.0163 $\pm$ 0.0085	0.0083 $\pm$ 0.0036	0.0117 $\pm$ 0.0101	0.0092 $\pm$ 0.0090
		ASPD $\downarrow$	0.0577 $\pm$ 0.0113	0.0598 $\pm$ 0.0107	0.0398 $\pm$ 0.0179	0.0431 $\pm$ 0.0132	0.0532 $\pm$ 0.0138	0.0561 $\pm$ 0.0108	0.0642 $\pm$ 0.0094	0.0430 $\pm$ 0.0106	0.0472 $\pm$ 0.0108	0.0505 $\pm$ 0.0089
Robust FL	FLTrust	ACC (%) $\uparrow$	89.13 $\pm$ 0.84	89.35 $\pm$ 0.92	89.27 $\pm$ 0.85	89.27 $\pm$ 0.85	89.29 $\pm$ 0.78	89.36 $\pm$ 0.63	89.55 $\pm$ 0.76	89.09 $\pm$ 0.72	89.09 $\pm$ 0.72	89.24 $\pm$ 0.74
		AEOD $\downarrow$	0.0086 $\pm$ 0.0065	0.0076 $\pm$ 0.0067	0.0074 $\pm$ 0.0061	0.0074 $\pm$ 0.0061	0.0127 $\pm$ 0.0112	0.0075 $\pm$ 0.0056	0.0122 $\pm$ 0.0072	0.0097 $\pm$ 0.0070	0.0097 $\pm$ 0.0070	0.0096 $\pm$ 0.0066
		ASPD $\downarrow$	0.0750 $\pm$ 0.0076	0.0743 $\pm$ 0.0116	0.0650 $\pm$ 0.0099	0.0650 $\pm$ 0.0099	0.0769 $\pm$ 0.0181	0.0738 $\pm$ 0.0076	0.0771 $\pm$ 0.0118	0.0681 $\pm$ 0.0116	0.0681 $\pm$ 0.0116	0.0691 $\pm$ 0.0101
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.05 $\pm$ 1.44	88.62 $\pm$ 1.12	88.49 $\pm$ 1.22	88.46 $\pm$ 1.22	88.18 $\pm$ 1.02	88.24 $\pm$ 1.16	88.12 $\pm$ 1.30	88.28 $\pm$ 1.27	88.16 $\pm$ 1.54	88.07 $\pm$ 1.26
		AEOD $\downarrow$	0.0140 $\pm$ 0.0105	0.0124 $\pm$ 0.0103	0.0130 $\pm$ 0.0131	0.0113 $\pm$ 0.0116	0.0144 $\pm$ 0.0093	0.0090 $\pm$ 0.0047	0.0133 $\pm$ 0.0078	0.0095 $\pm$ 0.0076	0.0123 $\pm$ 0.0066	0.0117 $\pm$ 0.0084
		ASPD $\downarrow$	0.0572 $\pm$ 0.0224	0.0566 $\pm$ 0.0222	0.0506 $\pm$ 0.0211	0.0574 $\pm$ 0.0187	0.0547 $\pm$ 0.0203	0.0595 $\pm$ 0.0137	0.0572 $\pm$ 0.0219	0.0537 $\pm$ 0.0158	0.0568 $\pm$ 0.0165	0.0535 $\pm$ 0.0225
Robust FL	LASA*	ACC (%) $\uparrow$	88.97 $\pm$ 1.40	89.00 $\pm$ 0.83	85.45 $\pm$ 1.71	85.64 $\pm$ 1.62	87.80 $\pm$ 1.39	88.92 $\pm$ 0.88	89.14 $\pm$ 0.86	85.50 $\pm$ 1.65	85.59 $\pm$ 1.57	87.77 $\pm$ 1.19
		AEOD $\downarrow$	0.0095 $\pm$ 0.0074	0.0086 $\pm$ 0.0070	0.0098 $\pm$ 0.0070	0.0049 $\pm$ 0.0030	0.0078 $\pm$ 0.0044	0.0101 $\pm$ 0.0050	0.0076 $\pm$ 0.0057	0.0060 $\pm$ 0.0047	0.0086 $\pm$ 0.0067	0.0047 $\pm$ 0.0056
		ASPD $\downarrow$	0.0662 $\pm$ 0.0126	0.0725 $\pm$ 0.0066	0.0483 $\pm$ 0.0090	0.0455 $\pm$ 0.0086	0.0590 $\pm$ 0.0099	0.0691 $\pm$ 0.0132	0.0738 $\pm$ 0.0046	0.0437 $\pm$ 0.0075	0.0453 $\pm$ 0.0106	0.0646 $\pm$ 0.0080
Robust + fair	FairGuard*	ACC (%) $\uparrow$	88.92 $\pm$ 0.91	89.18 $\pm$ 1.24	51.67 $\pm$ 0.01	51.67 $\pm$ 0.00	83.15 $\pm$ 1.26	89.09 $\pm$ 1.05	89.55 $\pm$ 0.66	61.22 $\pm$ 15.39	62.27 $\pm$ 14.60	78.33 $\pm$ 14.24
		AEOD $\downarrow$	0.0086 $\pm$ 0.0049	0.0079 $\pm$ 0.0054	0.0002 $\pm$ 0.0003	0.0000 $\pm$ 0.0001	0.0101 $\pm$ 0.0059	0.0105 $\pm$ 0.0085	0.0153 $\pm$ 0.0095	0.0051 $\pm$ 0.0099	0.0066 $\pm$ 0.0086	0.0127 $\pm$ 0.0102
		ASPD $\downarrow$	0.0593 $\pm$ 0.0118	0.0676 $\pm$ 0.0158	0.0001 $\pm$ 0.0002	0.0000 $\pm$ 0.0001	0.0221 $\pm$ 0.0078	0.0597 $\pm$ 0.0141	0.0725 $\pm$ 0.0175	0.0044 $\pm$ 0.0133	0.0048 $\pm$ 0.0099	0.0191 $\pm$ 0.0168
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.60 $\pm$ 1.11	89.98 $\pm$ 0.66	80.70 $\pm$ 6.93	76.87 $\pm$ 14.90	86.81 $\pm$ 3.77	89.31 $\pm$ 1.62	89.73 $\pm$ 1.31	81.54 $\pm$ 11.50	80.46 $\pm$ 10.78	81.23 $\pm$ 11.08
		AEOD $\downarrow$	0.0093 $\pm$ 0.0071	0.0085 $\pm$ 0.0066	0.0104 $\pm$ 0.0111	0.0107 $\pm$ 0.0082	0.0079 $\pm$ 0.0045	0.0111 $\pm$ 0.0083	0.0066 $\pm$ 0.0053	0.0132 $\pm$ 0.0102	0.0125 $\pm$ 0.0099	0.0136 $\pm$ 0.0135
		ASPD $\downarrow$	0.0689 $\pm$ 0.0168	0.0707 $\pm$ 0.0145	0.0196 $\pm$ 0.0242	0.0255 $\pm$ 0.0216	0.0508 $\pm$ 0.0205	0.0675 $\pm$ 0.0122	0.0697 $\pm$ 0.0112	0.0369 $\pm$ 0.0299	0.0273 $\pm$ 0.0203	0.0414 $\pm$ 0.0212
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.26 $\pm$ 1.04	88.39 $\pm$ 0.63	88.47 $\pm$ 0.91	88.69 $\pm$ 0.81	88.38 $\pm$ 0.89	88.59 $\pm$ 1.17	88.44 $\pm$ 0.74	88.41 $\pm$ 0.92	88.22 $\pm$ 0.78	88.34 $\pm$ 0.96
		AEOD $\downarrow$	0.0097 $\pm$ 0.0075	0.0107 $\pm$ 0.0096	0.0061 $\pm$ 0.0042	0.0077 $\pm$ 0.0046	0.0165 $\pm$ 0.0072	0.0070 $\pm$ 0.0042	0.0100 $\pm$ 0.0070	0.0078 $\pm$ 0.0056	0.0126 $\pm$ 0.0093	0.0087 $\pm$ 0.0066
		ASPD $\downarrow$	0.0625 $\pm$ 0.0138	0.0607 $\pm$ 0.0103	0.0640 $\pm$ 0.0142	0.0638 $\pm$ 0.0070	0.0559 $\pm$ 0.0156	0.0651 $\pm$ 0.0090	0.0589 $\pm$ 0.0092	0.0616 $\pm$ 0.0141	0.0584 $\pm$ 0.0189	0.0596 $\pm$ 0.0146

All displayed cells use the same 10 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
This table uses 886 cu128 and 14 cu130 records; the source cohort contains 14 cu130 records. No 70-round equivalence claim.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Shared-calibration view: all nine methods use saved clean-training-root group thresholds; no validation-label fitting.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Shared calibration

9 shared seeds: 91002–91010; round 70; validation split (19,867 images)

Table 8: CelebA [shared calibration]: IID / non-IID. COMPLETED VALIDATION COHORT – 9 shared seeds (91002–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	88.02 $\pm$ 1.84	89.37 $\pm$ 0.68	70.08 $\pm$ 4.92	67.33 $\pm$ 2.98	84.57 $\pm$ 1.66	87.74 $\pm$ 4.05	89.48 $\pm$ 1.45	75.90 $\pm$ 12.95	75.47 $\pm$ 13.23	85.18 $\pm$ 2.61
		AEOD $\downarrow$	0.0068 $\pm$ 0.0057	0.0100 $\pm$ 0.0079	0.0115 $\pm$ 0.0084	0.0110 $\pm$ 0.0081	0.0117 $\pm$ 0.0064	0.0085 $\pm$ 0.0053	0.0074 $\pm$ 0.0068	0.0107 $\pm$ 0.0070	0.0113 $\pm$ 0.0077	0.0163 $\pm$ 0.0119
		ASPD $\downarrow$	0.0607 $\pm$ 0.0172	0.0720 $\pm$ 0.0129	0.0092 $\pm$ 0.0071	0.0143 $\pm$ 0.0129	0.0299 $\pm$ 0.0183	0.0636 $\pm$ 0.0203	0.0719 $\pm$ 0.0156	0.0121 $\pm$ 0.0114	0.0179 $\pm$ 0.0124	0.0309 $\pm$ 0.0217
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.25 $\pm$ 0.75	89.50 $\pm$ 0.69	70.77 $\pm$ 3.91	64.24 $\pm$ 3.93	85.78 $\pm$ 1.25	89.42 $\pm$ 0.89	89.65 $\pm$ 0.97	72.49 $\pm$ 11.74	68.54 $\pm$ 11.62	86.14 $\pm$ 2.25
		AEOD $\downarrow$	0.0092 $\pm$ 0.0070	0.0070 $\pm$ 0.0076	0.0086 $\pm$ 0.0070	0.0114 $\pm$ 0.0104	0.0095 $\pm$ 0.0082	0.0084 $\pm$ 0.0051	0.0093 $\pm$ 0.0067	0.0092 $\pm$ 0.0077	0.0082 $\pm$ 0.0069	0.0107 $\pm$ 0.0033
		ASPD $\downarrow$	0.0707 $\pm$ 0.0094	0.0757 $\pm$ 0.0055	0.0077 $\pm$ 0.0061	0.0137 $\pm$ 0.0054	0.0410 $\pm$ 0.0117	0.0710 $\pm$ 0.0088	0.0766 $\pm$ 0.0087	0.0051 $\pm$ 0.0051	0.0090 $\pm$ 0.0075	0.0547 $\pm$ 0.0174
Robust FL	Median	ACC (%) $\uparrow$	86.08 $\pm$ 1.21	86.40 $\pm$ 1.33	84.53 $\pm$ 1.47	84.82 $\pm$ 1.17	85.38 $\pm$ 1.49	85.99 $\pm$ 1.22	86.09 $\pm$ 1.32	84.19 $\pm$ 1.31	84.25 $\pm$ 1.39	85.16 $\pm$ 1.33
		AEOD $\downarrow$	0.0121 $\pm$ 0.0070	0.0128 $\pm$ 0.0082	0.0131 $\pm$ 0.0128	0.0085 $\pm$ 0.0085	0.0136 $\pm$ 0.0077	0.0109 $\pm$ 0.0075	0.0170 $\pm$ 0.0088	0.0080 $\pm$ 0.0037	0.0103 $\pm$ 0.0097	0.0095 $\pm$ 0.0095
		ASPD $\downarrow$	0.0586 $\pm$ 0.0115	0.0600 $\pm$ 0.0113	0.0412 $\pm$ 0.0184	0.0450 $\pm$ 0.0125	0.0543 $\pm$ 0.0141	0.0569 $\pm$ 0.0111	0.0641 $\pm$ 0.0100	0.0422 $\pm$ 0.0109	0.0460 $\pm$ 0.0108	0.0509 $\pm$ 0.0094
Robust FL	FLTrust	ACC (%) $\uparrow$	89.18 $\pm$ 0.87	89.42 $\pm$ 0.95	89.32 $\pm$ 0.88	89.32 $\pm$ 0.88	89.38 $\pm$ 0.77	89.46 $\pm$ 0.58	89.61 $\pm$ 0.78	89.18 $\pm$ 0.70	89.18 $\pm$ 0.70	89.26 $\pm$ 0.79
		AEOD $\downarrow$	0.0085 $\pm$ 0.0068	0.0081 $\pm$ 0.0069	0.0080 $\pm$ 0.0062	0.0080 $\pm$ 0.0062	0.0137 $\pm$ 0.0114	0.0080 $\pm$ 0.0056	0.0130 $\pm$ 0.0072	0.0095 $\pm$ 0.0074	0.0095 $\pm$ 0.0074	0.0099 $\pm$ 0.0069
		ASPD $\downarrow$	0.0756 $\pm$ 0.0078	0.0750 $\pm$ 0.0120	0.0653 $\pm$ 0.0105	0.0653 $\pm$ 0.0105	0.0781 $\pm$ 0.0188	0.0749 $\pm$ 0.0071	0.0775 $\pm$ 0.0125	0.0700 $\pm$ 0.0105	0.0700 $\pm$ 0.0105	0.0686 $\pm$ 0.0106
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	88.30 $\pm$ 1.28	88.83 $\pm$ 0.97	88.54 $\pm$ 1.28	88.53 $\pm$ 1.28	88.42 $\pm$ 0.73	88.29 $\pm$ 1.22	88.28 $\pm$ 1.27	88.50 $\pm$ 1.14	88.52 $\pm$ 1.11	88.14 $\pm$ 1.32
		AEOD $\downarrow$	0.0139 $\pm$ 0.0112	0.0099 $\pm$ 0.0069	0.0143 $\pm$ 0.0131	0.0108 $\pm$ 0.0122	0.0150 $\pm$ 0.0096	0.0083 $\pm$ 0.0044	0.0133 $\pm$ 0.0082	0.0105 $\pm$ 0.0074	0.0129 $\pm$ 0.0067	0.0114 $\pm$ 0.0089
		ASPD $\downarrow$	0.0572 $\pm$ 0.0237	0.0613 $\pm$ 0.0172	0.0485 $\pm$ 0.0213	0.0594 $\pm$ 0.0187	0.0547 $\pm$ 0.0215	0.0569 $\pm$ 0.0117	0.0553 $\pm$ 0.0223	0.0550 $\pm$ 0.0162	0.0601 $\pm$ 0.0136	0.0513 $\pm$ 0.0226
Robust FL	LASA*	ACC (%) $\uparrow$	89.35 $\pm$ 0.77	89.13 $\pm$ 0.77	85.86 $\pm$ 1.21	85.81 $\pm$ 1.61	88.00 $\pm$ 1.31	89.12 $\pm$ 0.64	89.31 $\pm$ 0.71	85.91 $\pm$ 1.06	85.96 $\pm$ 1.09	87.94 $\pm$ 1.13
		AEOD $\downarrow$	0.0094 $\pm$ 0.0078	0.0070 $\pm$ 0.0053	0.0085 $\pm$ 0.0062	0.0054 $\pm$ 0.0027	0.0084 $\pm$ 0.0044	0.0093 $\pm$ 0.0046	0.0069 $\pm$ 0.0056	0.0066 $\pm$ 0.0046	0.0085 $\pm$ 0.0071	0.0030 $\pm$ 0.0018
		ASPD $\downarrow$	0.0678 $\pm$ 0.0124	0.0724 $\pm$ 0.0070	0.0478 $\pm$ 0.0094	0.0461 $\pm$ 0.0089	0.0592 $\pm$ 0.0105	0.0721 $\pm$ 0.0097	0.0738 $\pm$ 0.0049	0.0451 $\pm$ 0.0064	0.0466 $\pm$ 0.0103	0.0641 $\pm$ 0.0084
Robust + fair	FairGuard*	ACC (%) $\uparrow$	89.03 $\pm$ 0.89	89.25 $\pm$ 1.30	51.67 $\pm$ 0.01	51.67 $\pm$ 0.00	83.18 $\pm$ 1.33	89.23 $\pm$ 1.01	89.64 $\pm$ 0.63	62.28 $\pm$ 15.93	63.45 $\pm$ 14.97	77.68 $\pm$ 14.95
		AEOD $\downarrow$	0.0081 $\pm$ 0.0048	0.0080 $\pm$ 0.0058	0.0002 $\pm$ 0.0003	0.0000 $\pm$ 0.0000	0.0112 $\pm$ 0.0052	0.0108 $\pm$ 0.0089	0.0132 $\pm$ 0.0071	0.0057 $\pm$ 0.0104	0.0074 $\pm$ 0.0088	0.0128 $\pm$ 0.0109
		ASPD $\downarrow$	0.0605 $\pm$ 0.0118	0.0672 $\pm$ 0.0167	0.0001 $\pm$ 0.0002	0.0000 $\pm$ 0.0001	0.0215 $\pm$ 0.0080	0.0605 $\pm$ 0.0147	0.0695 $\pm$ 0.0156	0.0049 $\pm$ 0.0140	0.0053 $\pm$ 0.0103	0.0180 $\pm$ 0.0175
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.88 $\pm$ 0.69	90.05 $\pm$ 0.65	79.98 $\pm$ 6.94	75.58 $\pm$ 15.20	86.82 $\pm$ 3.99	89.81 $\pm$ 0.37	89.81 $\pm$ 1.36	81.04 $\pm$ 12.08	79.93 $\pm$ 11.30	80.49 $\pm$ 11.48
		AEOD $\downarrow$	0.0099 $\pm$ 0.0072	0.0094 $\pm$ 0.0063	0.0077 $\pm$ 0.0074	0.0106 $\pm$ 0.0087	0.0082 $\pm$ 0.0047	0.0110 $\pm$ 0.0088	0.0052 $\pm$ 0.0028	0.0120 $\pm$ 0.0100	0.0111 $\pm$ 0.0093	0.0136 $\pm$ 0.0143
		ASPD $\downarrow$	0.0715 $\pm$ 0.0154	0.0712 $\pm$ 0.0153	0.0216 $\pm$ 0.0248	0.0235 $\pm$ 0.0220	0.0514 $\pm$ 0.0216	0.0685 $\pm$ 0.0124	0.0693 $\pm$ 0.0118	0.0399 $\pm$ 0.0302	0.0297 $\pm$ 0.0199	0.0385 $\pm$ 0.0203
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.35 $\pm$ 1.06	88.41 $\pm$ 0.66	88.62 $\pm$ 0.82	88.84 $\pm$ 0.68	88.55 $\pm$ 0.77	88.59 $\pm$ 1.24	88.50 $\pm$ 0.76	88.39 $\pm$ 0.97	88.24 $\pm$ 0.82	88.47 $\pm$ 0.94
		AEOD $\downarrow$	0.0097 $\pm$ 0.0080	0.0115 $\pm$ 0.0098	0.0054 $\pm$ 0.0039	0.0068 $\pm$ 0.0038	0.0171 $\pm$ 0.0074	0.0063 $\pm$ 0.0039	0.0104 $\pm$ 0.0073	0.0083 $\pm$ 0.0057	0.0119 $\pm$ 0.0095	0.0079 $\pm$ 0.0065
		ASPD $\downarrow$	0.0623 $\pm$ 0.0146	0.0604 $\pm$ 0.0108	0.0667 $\pm$ 0.0120	0.0633 $\pm$ 0.0072	0.0555 $\pm$ 0.0165	0.0638 $\pm$ 0.0086	0.0579 $\pm$ 0.0090	0.0609 $\pm$ 0.0148	0.0605 $\pm$ 0.0188	0.0613 $\pm$ 0.0144

All displayed cells use the same 9 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Shared-calibration view: all nine methods use saved clean-training-root group thresholds; no validation-label fitting.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.

Shared calibration

6 shared seeds: 91005–91010; round 70; validation split (19,867 images)

Table 9: CelebA [shared calibration]: IID / non-IID. COMPLETED VALIDATION COHORT – 6 shared seeds (91005–91010). Validation-only; mean $\pm$ sample SD at round 70.

Category	Method	Metric	IID: Benign	IID: F-Flip	IID: FedSA	IID: S-DFA	IID: Sp-DFA	non-IID: Benign	non-IID: F-Flip	non-IID: FedSA	non-IID: S-DFA	non-IID: Sp-DFA
Vanilla FL	FedAvg	ACC (%) $\uparrow$	87.65 $\pm$ 2.13	89.43 $\pm$ 0.82	70.09 $\pm$ 5.93	67.79 $\pm$ 3.29	85.29 $\pm$ 1.34	86.58 $\pm$ 4.61	89.12 $\pm$ 1.66	78.18 $\pm$ 11.50	78.31 $\pm$ 12.23	85.62 $\pm$ 3.01
		AEOD $\downarrow$	0.0082 $\pm$ 0.0064	0.0091 $\pm$ 0.0067	0.0134 $\pm$ 0.0095	0.0117 $\pm$ 0.0101	0.0118 $\pm$ 0.0074	0.0086 $\pm$ 0.0062	0.0051 $\pm$ 0.0054	0.0116 $\pm$ 0.0069	0.0142 $\pm$ 0.0074	0.0131 $\pm$ 0.0102
		ASPD $\downarrow$	0.0587 $\pm$ 0.0200	0.0766 $\pm$ 0.0103	0.0094 $\pm$ 0.0069	0.0168 $\pm$ 0.0144	0.0334 $\pm$ 0.0172	0.0584 $\pm$ 0.0233	0.0650 $\pm$ 0.0105	0.0143 $\pm$ 0.0128	0.0230 $\pm$ 0.0116	0.0373 $\pm$ 0.0216
Fairness-aware	FairFed*	ACC (%) $\uparrow$	89.06 $\pm$ 0.86	89.28 $\pm$ 0.75	70.99 $\pm$ 4.24	65.02 $\pm$ 4.68	85.50 $\pm$ 1.19	89.20 $\pm$ 1.03	89.46 $\pm$ 1.16	74.79 $\pm$ 10.95	70.75 $\pm$ 12.06	85.85 $\pm$ 2.71
		AEOD $\downarrow$	0.0098 $\pm$ 0.0076	0.0086 $\pm$ 0.0088	0.0066 $\pm$ 0.0068	0.0148 $\pm$ 0.0110	0.0070 $\pm$ 0.0050	0.0070 $\pm$ 0.0054	0.0103 $\pm$ 0.0073	0.0120 $\pm$ 0.0076	0.0088 $\pm$ 0.0078	0.0117 $\pm$ 0.0024
		ASPD $\downarrow$	0.0727 $\pm$ 0.0107	0.0763 $\pm$ 0.0056	0.0106 $\pm$ 0.0053	0.0141 $\pm$ 0.0061	0.0438 $\pm$ 0.0097	0.0710 $\pm$ 0.0098	0.0778 $\pm$ 0.0089	0.0047 $\pm$ 0.0030	0.0082 $\pm$ 0.0060	0.0530 $\pm$ 0.0204
Robust FL	Median	ACC (%) $\uparrow$	85.87 $\pm$ 1.05	86.20 $\pm$ 1.29	84.10 $\pm$ 0.87	84.39 $\pm$ 0.75	85.07 $\pm$ 1.17	85.75 $\pm$ 1.03	85.75 $\pm$ 1.25	84.13 $\pm$ 1.13	84.20 $\pm$ 1.18	85.31 $\pm$ 1.19
		AEOD $\downarrow$	0.0112 $\pm$ 0.0085	0.0099 $\pm$ 0.0082	0.0126 $\pm$ 0.0144	0.0056 $\pm$ 0.0046	0.0138 $\pm$ 0.0089	0.0084 $\pm$ 0.0068	0.0192 $\pm$ 0.0095	0.0071 $\pm$ 0.0022	0.0095 $\pm$ 0.0090	0.0120 $\pm$ 0.0109
		ASPD $\downarrow$	0.0559 $\pm$ 0.0115	0.0557 $\pm$ 0.0104	0.0341 $\pm$ 0.0158	0.0382 $\pm$ 0.0066	0.0529 $\pm$ 0.0116	0.0528 $\pm$ 0.0101	0.0658 $\pm$ 0.0099	0.0404 $\pm$ 0.0119	0.0434 $\pm$ 0.0104	0.0562 $\pm$ 0.0050
Robust FL	FLTrust	ACC (%) $\uparrow$	89.08 $\pm$ 0.92	89.42 $\pm$ 0.97	89.17 $\pm$ 1.00	89.17 $\pm$ 1.00	89.26 $\pm$ 0.84	89.26 $\pm$ 0.62	89.46 $\pm$ 0.93	89.11 $\pm$ 0.63	89.11 $\pm$ 0.63	89.20 $\pm$ 0.88
		AEOD $\downarrow$	0.0068 $\pm$ 0.0068	0.0081 $\pm$ 0.0087	0.0059 $\pm$ 0.0060	0.0059 $\pm$ 0.0060	0.0092 $\pm$ 0.0112	0.0099 $\pm$ 0.0061	0.0127 $\pm$ 0.0090	0.0102 $\pm$ 0.0078	0.0102 $\pm$ 0.0078	0.0102 $\pm$ 0.0077
		ASPD $\downarrow$	0.0746 $\pm$ 0.0060	0.0782 $\pm$ 0.0058	0.0673 $\pm$ 0.0105	0.0673 $\pm$ 0.0105	0.0765 $\pm$ 0.0142	0.0765 $\pm$ 0.0082	0.0800 $\pm$ 0.0115	0.0669 $\pm$ 0.0116	0.0669 $\pm$ 0.0116	0.0664 $\pm$ 0.0122
Adaptive FL	FedAA-DDPG*	ACC (%) $\uparrow$	87.86 $\pm$ 1.35	88.47 $\pm$ 1.01	87.99 $\pm$ 1.22	88.13 $\pm$ 1.36	88.05 $\pm$ 0.60	88.20 $\pm$ 1.48	88.04 $\pm$ 1.41	88.50 $\pm$ 1.37	88.41 $\pm$ 1.21	87.94 $\pm$ 1.59
		AEOD $\downarrow$	0.0142 $\pm$ 0.0105	0.0109 $\pm$ 0.0081	0.0168 $\pm$ 0.0147	0.0131 $\pm$ 0.0139	0.0133 $\pm$ 0.0081	0.0095 $\pm$ 0.0037	0.0146 $\pm$ 0.0100	0.0091 $\pm$ 0.0085	0.0109 $\pm$ 0.0066	0.0120 $\pm$ 0.0088
		ASPD $\downarrow$	0.0569 $\pm$ 0.0289	0.0549 $\pm$ 0.0172	0.0415 $\pm$ 0.0212	0.0531 $\pm$ 0.0193	0.0563 $\pm$ 0.0258	0.0570 $\pm$ 0.0120	0.0497 $\pm$ 0.0219	0.0545 $\pm$ 0.0168	0.0535 $\pm$ 0.0079	0.0479 $\pm$ 0.0262
Robust FL	LASA*	ACC (%) $\uparrow$	89.11 $\pm$ 0.85	88.76 $\pm$ 0.69	85.31 $\pm$ 1.08	85.25 $\pm$ 1.66	87.47 $\pm$ 1.32	88.89 $\pm$ 0.67	89.26 $\pm$ 0.89	85.65 $\pm$ 1.15	85.69 $\pm$ 1.22	87.69 $\pm$ 1.31
		AEOD $\downarrow$	0.0113 $\pm$ 0.0086	0.0058 $\pm$ 0.0028	0.0099 $\pm$ 0.0069	0.0056 $\pm$ 0.0028	0.0086 $\pm$ 0.0048	0.0074 $\pm$ 0.0044	0.0084 $\pm$ 0.0062	0.0050 $\pm$ 0.0023	0.0075 $\pm$ 0.0055	0.0025 $\pm$ 0.0009
		ASPD $\downarrow$	0.0642 $\pm$ 0.0130	0.0710 $\pm$ 0.0044	0.0454 $\pm$ 0.0107	0.0433 $\pm$ 0.0062	0.0566 $\pm$ 0.0120	0.0688 $\pm$ 0.0065	0.0752 $\pm$ 0.0048	0.0448 $\pm$ 0.0055	0.0437 $\pm$ 0.0095	0.0630 $\pm$ 0.0104
Robust + fair	FairGuard*	ACC (%) $\uparrow$	88.79 $\pm$ 1.02	88.91 $\pm$ 1.50	51.67 $\pm$ 0.00	51.67 $\pm$ 0.00	82.89 $\pm$ 1.55	88.83 $\pm$ 1.01	89.52 $\pm$ 0.72	62.12 $\pm$ 16.21	63.84 $\pm$ 14.59	79.68 $\pm$ 13.77
		AEOD $\downarrow$	0.0091 $\pm$ 0.0046	0.0084 $\pm$ 0.0072	0.0001 $\pm$ 0.0002	0.0000 $\pm$ 0.0000	0.0115 $\pm$ 0.0047	0.0106 $\pm$ 0.0103	0.0138 $\pm$ 0.0077	0.0077 $\pm$ 0.0124	0.0086 $\pm$ 0.0094	0.0159 $\pm$ 0.0118
		ASPD $\downarrow$	0.0569 $\pm$ 0.0131	0.0635 $\pm$ 0.0186	0.0001 $\pm$ 0.0002	0.0000 $\pm$ 0.0000	0.0202 $\pm$ 0.0084	0.0562 $\pm$ 0.0164	0.0654 $\pm$ 0.0151	0.0003 $\pm$ 0.0004	0.0028 $\pm$ 0.0043	0.0158 $\pm$ 0.0211
Robust + fair	FLTrust+FairGuard*	ACC (%) $\uparrow$	89.63 $\pm$ 0.68	89.97 $\pm$ 0.80	82.00 $\pm$ 7.22	82.79 $\pm$ 11.90	88.48 $\pm$ 1.33	89.99 $\pm$ 0.22	89.81 $\pm$ 1.71	84.01 $\pm$ 5.97	84.31 $\pm$ 3.36	84.30 $\pm$ 9.79
		AEOD $\downarrow$	0.0109 $\pm$ 0.0081	0.0109 $\pm$ 0.0069	0.0050 $\pm$ 0.0047	0.0130 $\pm$ 0.0094	0.0067 $\pm$ 0.0046	0.0145 $\pm$ 0.0086	0.0051 $\pm$ 0.0032	0.0144 $\pm$ 0.0111	0.0108 $\pm$ 0.0099	0.0081 $\pm$ 0.0062
		ASPD $\downarrow$	0.0662 $\pm$ 0.0135	0.0687 $\pm$ 0.0182	0.0302 $\pm$ 0.0266	0.0312 $\pm$ 0.0231	0.0609 $\pm$ 0.0125	0.0686 $\pm$ 0.0157	0.0693 $\pm$ 0.0148	0.0431 $\pm$ 0.0291	0.0319 $\pm$ 0.0202	0.0469 $\pm$ 0.0169
Ours	GuardFed-AD2+	ACC (%) $\uparrow$	88.56 $\pm$ 0.68	88.29 $\pm$ 0.75	88.57 $\pm$ 0.87	88.75 $\pm$ 0.79	88.35 $\pm$ 0.87	88.58 $\pm$ 1.42	88.43 $\pm$ 0.87	88.49 $\pm$ 0.88	88.42 $\pm$ 0.83	88.42 $\pm$ 1.10
		AEOD $\downarrow$	0.0114 $\pm$ 0.0073	0.0148 $\pm$ 0.0104	0.0055 $\pm$ 0.0036	0.0070 $\pm$ 0.0040	0.0171 $\pm$ 0.0083	0.0062 $\pm$ 0.0039	0.0110 $\pm$ 0.0086	0.0083 $\pm$ 0.0044	0.0067 $\pm$ 0.0068	0.0089 $\pm$ 0.0080
		ASPD $\downarrow$	0.0665 $\pm$ 0.0159	0.0581 $\pm$ 0.0129	0.0697 $\pm$ 0.0110	0.0617 $\pm$ 0.0085	0.0559 $\pm$ 0.0203	0.0641 $\pm$ 0.0067	0.0570 $\pm$ 0.0090	0.0566 $\pm$ 0.0081	0.0611 $\pm$ 0.0114	0.0583 $\pm$ 0.0172

All displayed cells use the same 6 seeds. Seed subsets follow the frozen protocol; all cells within each table use identical seeds.
Round 70; validation only (19,867 images). Mean +/- sample SD (ddof = 1); ACC in %, gaps on [0, 1].
AEOD is the implemented absolute TPR gap. Descriptive values do not establish statistical significance.
* Adaptations: FairFed, FairGuard, hybrid; FedAA-DDPG round/policy adapter; LASA with local-Adam update differences.
Recipes fixed before coverage. Seven methods used non-IID Benign/S-DFA search; FedAA/LASA used both distributions, Benign/S-DFA, seed 91001.
All records in this table use torch 2.11/cu128. The source cohort contains 14 cu130 records, excluded from this table.
Nine methods only; eight additional manuscript baselines and final frozen evaluation remain. Native/shared main endpoint is pending.
Shared-calibration view: all nine methods use saved clean-training-root group thresholds; no validation-label fitting.
All outcomes, including constant predictions, are retained. Inference: 434 CPU / 466 GPU in the 900-ID source; no uniform-device claim.

Decision rules. Raw: margin > 0 (ties class 0). Native: GuardFed-AD2+ uses its saved clean-root group thresholds; the eight baselines use argmax. Shared: all nine use saved clean-root group thresholds ($\geq$); no validation-label fitting.

Reading scope. Validation was exposed to recipe selection, including seed 91001. Mixed CPU/GPU replay is not a uniform-device final comparison. The native/shared primary endpoint remains pending.